

Reproducing Kernel Hilbert Spaces

Reading notes, Lutz Dümbgen, August 9, 2021

Main external sources:

M. Hein, O. Bousquet (2004): Kernels, associated structures and generalizations. Technical report 127, Max-Planck-Institut für biologische Kybernetik, Tübingen.

K. Muandet, K. Fukumizu, B. Sriperumbudur, B. Schölkopf (2020): *Kernel Mean Embedding of Distributions: A Review and Beyond*. arXiv:1605.09522v4.

D. Ginsbourger (2021): Lecture notes on Reproducing Kernel Hilbert Spaces.

F. Tomi (1987): Lecture notes on Functional Analysis.

(Plus references cited therein.)

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1 From Hilbert function spaces to positive definite kernels

In what follows, let $\mathcal{X}$ be an arbitrary set, and let $\mathcal{H} = (\mathcal{H}, \langle \cdot, \cdot \rangle, \| \cdot \|)$ be a real Hilbert space of functions $f : \mathcal{X} \rightarrow \mathbb{R}$.

Definition 1.1 (Positive definite kernel). A *positive definite kernel* on $\mathcal{X}$ is a symmetric function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ such that for arbitrary $m \in \mathbb{N}$, points $x_1, \dots, x_m \in \mathcal{X}$ and scalars $c_1, \dots, c_m \in \mathbb{R}$,

$$\sum_{i,j=1}^m c_i c_j k(x_i, x_j) \geq 0.$$

A positive definite kernel k on $\mathcal{X}$ is called *strictly positive definite*, if the previous inequality is strict whenever $x_1, \dots, x_m$ are pairwise different and $c_1, \dots, c_m \neq 0$.

The attribute “positive definite” is a bit misleading. One could also say “positive semidefinite” or “nonnegative definite”. Indeed, a function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a positive definite kernel if and only if for arbitrary $m \in \mathbb{N}$ and points $x_1, \dots, x_m \in \mathcal{X}$, the so-called Gram matrix $(k(x_i, x_j))_{i,j=1}^m \in \mathbb{R}^{m \times m}$ is symmetric and positive semidefinite. A particular consequence of k being positive definite is that

$$k(x, x) \geq 0 \quad \text{and} \quad |k(x, y)| \leq \sqrt{k(x, x)k(y, y)}$$

for arbitrary $x, y \in \mathcal{X}$.

A function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a strictly positive definite kernel if and only if for arbitrary $m \in \mathbb{N}$ and pairwise different points $x_1, \dots, x_m \in \mathcal{X}$, the gram matrix $(k(x_i, x_j))_{i,j=1}^m$ is symmetric and positive definite. A particular consequence of k being strictly positive definite is that $k(x, x) > 0$ for all $x \in \mathcal{X}$, and $|k(x, y)| < \sqrt{k(x, x)k(y, y)}$ for arbitrary different points $x, y \in \mathcal{X}$.

Theorem 1.2. *The following two properties of $\mathcal{H}$ are equivalent:*

- (a) *For each $x \in \mathcal{X}$, the mapping $\mathcal{H} \ni f \mapsto f(x) \in \mathbb{R}$ is continuous.*
- (b) *There exists a function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ satisfying the following two conditions:*

- (b.1) $k(x, \cdot) \in \mathcal{H}$ for all $x \in \mathcal{X}$;
- (b.2) $\langle f, k(x, \cdot) \rangle = f(x)$ for all $x \in \mathcal{X}$ and any $f \in \mathcal{H}$.

The function k in property (b) is unique and defines a positive definite kernel on $\mathcal{X}$. Furthermore, the linear span of all functions $k(x, \cdot)$, $x \in \mathcal{X}$, is dense in $\mathcal{H}$.

Definition 1.3 (Reproducing kernel Hilbert space). A real Hilbert function space $\mathcal{H}$ with properties (a–b) in Theorem 1.2 is called *reproducing kernel Hilbert space (RKHS)*, and the function k in property (b) is called the *reproducing kernel of $\mathcal{H}$* .

Proof of Theorem 1.2. Suppose that $\mathcal{H}$ has property (a). Then it follows from *Riesz’ representation theorem* that for each $x \in \mathcal{X}$ there exists a function $R_x \in \mathcal{H}$ such that $f(x) = \langle f, R_x \rangle$ for all $f \in \mathcal{H}$. If we define $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ via $k(x, y) := R_x(y)$, then $k(x, \cdot) = R_x \in \mathcal{H}$ for

any $x \in \mathcal{X}$, and $\langle f, k(x, \cdot) \rangle = \langle f, R_x \rangle = f(x)$ for arbitrary $x \in \mathcal{X}$ and $f \in \mathcal{H}$. Hence, $\mathcal{H}$ has property (b) as well.

Suppose that $\mathcal{H}$ has property (b). Then for each $x \in \mathcal{X}$, the Cauchy-Schwarz inequality implies that $f(x) = \langle f, k(x, \cdot) \rangle$ satisfies the inequality $|f(x)| \leq \|k(x, \cdot)\| \|f\|$. Thus for any fixed $x \in \mathcal{X}$, $f \mapsto f(x)$ defines a continuous linear functional on $\mathcal{H}$, i.e. $\mathcal{H}$ has property (a).

Concerning uniqueness and symmetry of k in condition (b), suppose that k' and k'' are two functions from $\mathcal{X} \times \mathcal{X}$ to $\mathbb{R}$ satisfying (b.1–2). Then for arbitrary $x, y \in \mathcal{X}$,

$$\begin{aligned} k'(x, y) &= \langle k'(x, \cdot), k''(y, \cdot) \rangle && \text{((b.1) for } k', \text{(b.2) for } k'') \\ &= \langle k''(y, \cdot), k'(x, \cdot) \rangle && \text{(symmetry of } \langle \cdot, \cdot \rangle) \\ &= k''(y, x) && \text{((b.1) for } k'', \text{(b.2) for } k'). \end{aligned}$$

In the special case of $k' \equiv k''$, this shows that any function k satisfying (b.1–2) has to be symmetric. Then for arbitrary functions k', k'' satisfying (b.1–2), we obtain uniqueness.

Positive definiteness of k follows from the fact that $f := \sum_{i=1}^m c_i k(x_i, \cdot)$ defines a function in $\mathcal{H}$, and

$$0 \leq \|f\|^2 = \langle f, f \rangle = \sum_{i,j=1}^m c_i c_j \langle k(x_i, \cdot), k(x_j, \cdot) \rangle = \sum_{i,j=1}^m c_i c_j k(x_i, x_j)$$

by property (b.2).

Finally, let $f \in \mathcal{H}$ be perpendicular to the closure of $\text{span}\{k(x, \cdot) : x \in \mathcal{X}\}$. This is equivalent to $\langle f, k(x, \cdot) \rangle = 0$ for all $x \in \mathcal{X}$. But $\langle f, k(x, \cdot) \rangle = f(x)$, so $f \equiv 0$. $\square$

2 From positive definite kernels to Hilbert function spaces

Now we start from an arbitrary set $\mathcal{X}$ and a symmetric, positive definite kernel k on $\mathcal{X}$. The question is whether there exists a RKHS $\mathcal{H}$ such that k is the corresponding reproducing kernel. The answer is yes!

Theorem 2.1 (Moore–Aronszajn). *For any symmetric, positive definite kernel $k(\cdot, \cdot)$ on $\mathcal{X}$ there exists a unique reproducing kernel Hilbert space $\mathcal{H}$ such that $k(\cdot, \cdot)$ is its reproducing kernel.*

Proof of Theorem 2.1. We consider the linear space $\mathcal{H}_o := \text{span}\{k(x, \cdot) : x \in \mathcal{X}\}$. For arbitrary functions $f = \sum_{i=1}^m c_i k(x_i, \cdot)$ and $g = \sum_{j=1}^n d_j k(y_j, \cdot)$ in $\mathcal{H}_o$, we set

$$\langle f, g \rangle := \sum_{i=1}^m \sum_{j=1}^n c_i d_j k(x_i, y_j) = \begin{cases} \sum_{i=1}^m c_i g(x_i), \\ \sum_{j=1}^n d_j f(y_j). \end{cases}$$

The latter two expressions show that the definition of $\langle f, g \rangle$ does not depend on the particular choice of the representations of f and g . This defines a symmetric bilinear form $\langle \cdot, \cdot \rangle$ on $\mathcal{H}_o$, and

for any $f \in \mathcal{H}_o$ and $x \in \mathcal{X}$,

$$f(x) = \langle f, k(x, \cdot) \rangle.$$

Since k is positive definite, this bilinear form is positive semidefinite in the sense that $\langle f, f \rangle \geq 0$ for all $f \in \mathcal{H}_o$, and $\|f\| := \langle f, f \rangle^{1/2}$ defines a seminorm on $\mathcal{H}_o$ satisfying the Cauchy–Schwarz inequality,

$$\pm \langle f, g \rangle \leq \|f\| \|g\| \quad \text{for arbitrary } f, g \in \mathcal{H}_o.$$

In particular, for $f \in \mathcal{H}_o$ and $x \in \mathcal{X}$,

$$|f(x)| \leq \|k(x, \cdot)\| \|f\| = k(x, x)^{1/2} \|f\|.$$

Consequently, $\|\cdot\|$ is even a norm on $\mathcal{H}_o$ in the sense that $\|f\| = 0$ implies that $f \equiv 0$. This leads to a pre-Hilbert space $(\mathcal{H}_o, \langle \cdot, \cdot \rangle, \|\cdot\|)$.

In general, $\mathcal{H}_o$ is not complete. But if $(f_n)_{n \geq 1}$ is a Cauchy sequence in $(\mathcal{H}_o, \|\cdot\|)$, then for any point $x \in \mathcal{X}$ and $m, n \in \mathbb{N}$,

$$|f_n(x) - f_m(x)| \leq k(x, x)^{1/2} \|f_n - f_m\|,$$

whence $(f_n(x))_{n \geq 1}$ is a Cauchy sequence in $\mathbb{R}$. Denoting its limit with $f(x)$, we see that any Cauchy sequence $(f_n)_{n \geq 1}$ in $\mathcal{H}_o$ converges pointwise to a function $f : \mathcal{X} \rightarrow \mathbb{R}$. Moreover, if $(f_n)_{n \geq 1}$ and $(g_n)_{n \geq 1}$ are two Cauchy sequences in $(\mathcal{H}_o, \|\cdot\|)$ converging pointwise to the same function $f : \mathcal{X} \rightarrow \mathbb{R}$, then $\lim_{n \rightarrow \infty} \|f_n - g_n\| = 0$. To verify this claim, note that $h_n := f_n - g_n$ defines a Cauchy sequence in $(\mathcal{H}_o, \|\cdot\|)$ converging pointwise to 0. The Cauchy property is equivalent to $\delta_N := \sup_{n \geq N} \|h_n - h_N\| \rightarrow 0$ as $N \rightarrow \infty$. For fixed N , writing $h_N = \sum_{i=1}^m c_i k(x_i, \cdot)$, we may conclude that for $n \geq N$,

$$\begin{aligned} \delta(N)^2 &\geq \|h_N - h_n\|^2 \geq \|h_N\|^2 - 2\langle h_n, h_N \rangle \\ &= \|h_N\|^2 - 2 \sum_{i=1}^m c_i h_n(x_i) \\ &\rightarrow \|h_N\|^2 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Thus $\|h_N\| \leq \delta_N \rightarrow 0$ as $N \rightarrow \infty$.

Now let $\mathcal{H}$ be the set of all functions $f : \mathcal{X} \rightarrow \mathbb{R}$ which are the pointwise limit of a Cauchy sequence in $(\mathcal{H}_o, \|\cdot\|)$. Obviously, $\mathcal{H}_o \subset \mathcal{H}$. For $f, g \in \mathcal{H}$ with corresponding Cauchy sequences $(f_n)_{n \geq 1}$ and $(g_n)_{n \geq 1}$ in $\mathcal{H}_o$, we may define

$$\langle f, g \rangle := \lim_{n \rightarrow \infty} \langle f_n, g_n \rangle.$$

This limit is well-defined, because $C_f := \sup_{n \geq 1} \|f_n\|$ and $C_g := \sup_{n \geq 1} \|g_n\|$ are finite, so by the Cauchy–Schwarz inequality,

$$\sup_{m, n \geq N} |\langle f_n, g_n \rangle - \langle f_m, g_m \rangle| \leq \sup_{m, n \geq N} (C_g \|f_n - f_m\| + C_f \|g_n - g_m\|) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Hence, $(\langle f_n, g_n \rangle)_{n \geq 1}$ is a Cauchy sequence in $\mathbb{R}$. Moreover, the limit does not depend on the particular choice of the sequences $(f_n)_{n \geq 1}$ and $(g_n)_{n \geq 1}$. For instance, any other Cauchy sequence $(\tilde{f}_n)_{n \geq 1}$ with pointwise limit f would satisfy $\lim_{n \rightarrow \infty} \|\tilde{f}_n - f_n\| = 0$, whence

$$|\langle \tilde{f}_n, g_n \rangle - \langle f_n, g_n \rangle| \leq C_g \|\tilde{f}_n - f_n\| \rightarrow 0$$

as $n \rightarrow \infty$. Moreover, in the special case of $g_n = k(x, \cdot)$ for all $n \geq 1$ with a fixed $x \in \mathcal{X}$, we obtain the formula

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \langle f_n, k(x, \cdot) \rangle = \langle f, k(x, \cdot) \rangle.$$

One can easily verify that $\langle \cdot, \cdot \rangle$ defines an inner product on $\mathcal{H}$, that is, $\langle \cdot, \cdot \rangle$ is bilinear, and $\langle f, f \rangle \geq 0$ for all $f \in \mathcal{H}$ with equality if and only if $f \equiv 0$. It remains to prove that $\mathcal{H}$ is complete with respect to the norm $\|\cdot\|$. Note first that for $f \in \mathcal{H}$ and any Cauchy sequence $(f_n)_{n \geq 1}$ in $(\mathcal{H}_o, \|\cdot\|)$ with pointwise limit f ,

$$\begin{aligned} \|f - f_n\|^2 &= \langle f, f \rangle - 2\langle f, f_n \rangle + \langle f_n, f_n \rangle \\ &= \lim_{m \rightarrow \infty} (\langle f_m, f_m \rangle - 2\langle f_m, f_n \rangle + \langle f_n, f_n \rangle) \\ &\leq \sup_{m \geq n} \|f_m - f_n\|^2 \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Now let $(f_n)_{n \geq 1}$ be a Cauchy sequence in $(\mathcal{H}, \|\cdot\|)$. For each $n \geq 1$ there exists a function $\tilde{f}_n \in \mathcal{H}_o$ such that $\|f_n - \tilde{f}_n\| \leq n^{-1}$. But then, $(\tilde{f}_n)_{n \geq 1}$ is a Cauchy sequence in $(\mathcal{H}_o, \|\cdot\|)$, whence it converges pointwise to some function $f \in \mathcal{H}$. In particular, f is the limit of $(f_n)_{n \geq 1}$ in $(\mathcal{H}, \|\cdot\|)$, because $\|f - f_n\| \leq \|f - \tilde{f}_n\| + n^{-1} \rightarrow 0$ as $n \rightarrow \infty$.

Let us explain uniqueness of the RKHS $\mathcal{H}$. As shown in Theorem 1.2, any RKHS $\mathcal{H}$ with kernel k has to contain the linear space $\mathcal{H}_o = \text{span}(k(x, \cdot) : x \in \mathcal{X})$, and $\mathcal{H} = \overline{\mathcal{H}_o}$. For functions $f, g \in \mathcal{H}_o$, the inner product $\langle f, g \rangle$ has to coincide with the definition given above; this is just a consequence of the desired property (b) in Theorem 1.2. But then, since $\langle \cdot, \cdot \rangle$ is a continuous function on $\mathcal{H} \times \mathcal{H}$, and since any function in $\mathcal{H}$ is the limit of a Cauchy sequence in $\mathcal{H}_o$, there is no ambiguity about the function family $\mathcal{H}$ or the inner product $\langle \cdot, \cdot \rangle$. $\square$

3 Examples of (strictly) positive definite kernels

General recipe 1 (Feature maps). Let $x \mapsto \Phi_x$ be a mapping from $\mathcal{X}$ into a real pre-Hilbert space $(\mathcal{G}, \langle \cdot, \cdot \rangle, \|\cdot\|)$, a so-called feature map, and $\mathcal{G}$ is called a feature space. Then,

$$k(x, y) = \langle \Phi_x, \Phi_y \rangle \tag{3.1}$$

defines a positive definite kernel on $\mathcal{X}$. Indeed, symmetry of k follows from symmetry of the inner product, and for arbitrary $m \in \mathbb{N}$, points $x_1, \dots, x_m \in \mathcal{X}$ and scalars $c_1, \dots, c_m \in \mathbb{R}$,

$$\sum_{i,j=1}^m c_i c_j k(x_i, x_j) = \sum_{i,j=1}^m c_i c_j \langle \Phi_{x_i}, \Phi_{x_j} \rangle = \left\| \sum_{i=1}^m c_i \Phi_{x_i} \right\|^2 \geq 0.$$

If the mapping $x \mapsto \Phi_x$ has the additional property that $\Phi_{x_1}, \dots, \Phi_{x_m}$ are linearly independent whenever $x_1, \dots, x_m$ are pairwise different, then the kernel k is strictly positive definite.

This construction of a kernel can be reversed: To show that a given function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a positive definite kernel, one looks for a feature mapping $x \mapsto \Phi_x$ from $\mathcal{X}$ into a suitable feature space $(\mathcal{G}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ such that k is given by (3.1).

General recipe 2 (Embeddings). If $T : \mathcal{X} \rightarrow \mathcal{Y}$ is an arbitrary mapping, and if ℓ is a positive definite kernel on $\mathcal{Y}$, then

$$k(x, x') := \ell(T(x), T(x'))$$

defines a positive definite kernel on $\mathcal{X}$. If ℓ is strictly positive definite, and if T is injective, then k is strictly positive definite as well.

In particular, if k is a positive definite kernel on $\mathbb{R}^d$, then for any matrix $B \in \mathbb{R}^{d \times d}$, $k_B(x, y) := k(Bx, By)$ defines another positive definite kernel on $\mathbb{R}^d$. If k is strictly positive definite and B is nonsingular, then k_B is strictly positive definite too.

Example 3.1 (A Triangular kernel). For $x, y \in \mathbb{R}$ let

$$k(x, y) := (1 - |x - y|)^+.$$

This defines a strictly positive definite kernel k on $\mathbb{R}$. To see this, we identify any point $x \in \mathbb{R}$ with the function $\Phi_x \in L^2(\mathbb{R})$ given by

$$\Phi_x(t) := 1_{[x, x+1)}(t).$$

Then elementary calculations show that

$$k(x, y) = \int_{\mathbb{R}} \Phi_x(t) \Phi_y(t) dt = \langle \Phi_x, \Phi_y \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product of $L^2(\mathbb{R})$. Thus k is a positive definite kernel on $\mathbb{R}$.

To see that k is strictly positive definite, consider $m \in \mathbb{N}$, pairwise different points $x_1, \dots, x_m \in \mathbb{R}$ and real numbers $c_1, \dots, c_m \neq 0$. Without loss of generality let $x_1 < \dots < x_m$ in case of $m > 1$. Then $h := \sum_{i=1}^m c_i \Phi_{x_i}(t)$ satisfies the inequality $h(t)^2 \geq c_1^2 1_B(t)$ with the interval

$$B := \begin{cases} [x_1, x_1 + 1) & \text{if } m = 1, \\ [x_1, x_1 + 1] \setminus [x_2, \infty) & \text{if } m > 1, \end{cases}$$

whence $\int_{\mathbb{R}} h(t)^2 dt \geq c_1^2 \min(x_2 - x_1, 1) > 0$.

There is a multivariate generalization of this example: For $x, y \in \mathbb{R}^d$,

$$k(x, y) := \prod_{i=1}^d (1 - |x_i - y_i|)^+$$

defines a strictly positive definite kernel on $\mathbb{R}^d$. Indeed, if we identify $x \in \mathbb{R}^d$ with the indicator function $\Phi_x \in L^2(\mathbb{R}^d)$ of the d -dimensional cube

$$B_x := [x_1, x_1 + 1) \times \dots \times [x_d, x_d + 1),$$

then one can show that

$$k(x, y) = \int_{\mathbb{R}^d} \Phi_x(t) \Phi_y(t) dt = \text{Leb}(B_x \cap B_y).$$

This implies that k is positive definite. Strict positive definiteness can be verified as in the univariate case with the lexicographic order on $\mathbb{R}^d$.

Example 3.2 (The standard Gaussian kernel). Let $(\mathcal{X}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ be a real pre-Hilbert space. Then

$$k(x, y) := \exp(-\|x - y\|^2/2)$$

defines a strictly positive definite kernel on $\mathcal{X}$.

To see this, note that it suffices to show that k is a strictly positive definite kernel on any finite-dimensional linear subspace of $\mathcal{X}$. By choosing an orthonormal basis of such a space, we may identify it with $\mathbb{R}^d$ for some $d \in \mathbb{N}$, equipped with its standard inner product. Hence, it suffices to verify the claim in case of $\mathcal{X} = \mathbb{R}^d$. We identify an arbitrary point $x \in \mathbb{R}^d$ with the function $\Phi_x \in L^2(\mathbb{R}^d)$ given by

$$\Phi_x(t) := D \exp(-\|t - x\|^2)$$

with a constant $D > 0$. For $x, y \in \mathbb{R}^d$ let $z := 2^{-1}(x + y)$ and $\delta := 2^{-1}(x - y)$. Then $x = z + \delta$, $y = z - \delta$, and

$$\begin{aligned} \int_{\mathbb{R}^d} \Phi_x(t) \Phi_y(t) dt &= D^2 \int_{\mathbb{R}^d} \exp(-\|t - z - \delta\|^2 - \|t - z + \delta\|^2) dt \\ &= \exp(-2\|\delta\|^2) D^2 \int_{\mathbb{R}^d} \exp(-2\|t - z\|^2) dt \\ &= k(x, y) D^2 2^{-d/2} \int_{\mathbb{R}^d} \exp(-\|s\|^2) ds \\ &= k(x, y) D^2 (\pi/2)^{d/2}, \end{aligned}$$

where in the second to last step, $s = 2^{1/2}(t - z)$ and $dt = 2^{-d/2}ds$. Consequently, if we take $D = (2/\pi)^{d/4}$, then

$$k(x, y) = \langle \Phi_x, \Phi_y \rangle$$

for arbitrary $x, y \in \mathbb{R}^d$, where $\langle \cdot, \cdot \rangle$ denotes the inner product of $L^2(\mathbb{R}^d)$. This shows that k is a positive definite kernel on $\mathbb{R}^d$.

To see that k is strictly positive definite, consider $h := \sum_{i=1}^m c_i \Phi_{x_i}$ with $m \in \mathbb{N}$, pairwise different points $x_1, \dots, x_m \in \mathbb{R}^d$ and real numbers $c_1, \dots, c_m \neq 0$. We want to show that $\int_{\mathbb{R}^d} h(t)^2 dt > 0$. Since h is continuous, it suffices to show that $h \not\equiv 0$. For $m = 1$, this is obvious. In the case of $m > 1$, let

$$i_o \in \arg \max_{1 \leq i \leq m} \|x_i\|.$$

Then, $u_o := \|x_{i_o}\|^{-1} x_{i_o}$ defines a unit vector such that

$$x_i^\top u_o < x_{i_o}^\top u_o = \|x_{i_o}\| \quad \text{for } i \neq i_o.$$

Indeed, for $i \neq i_o$, let $x_i = \lambda u_o + v$ with $\lambda = x_i^\top u_o$ and $v \in u^\perp$. Then,

$$\|x_{i_o}\|^2 \geq \|x_i\|^2 = \lambda^2 + \|v\|^2,$$

so $|\lambda| \leq \|x_{i_o}\|$, and $\lambda = \|x_{i_o}\|$ would imply that $v = 0$, leading to the contradiction that $x_i = x_{i_o}$. Now the claim that $h \not\equiv 0$ follows by observing that for scalars $r > 0$,

$$\begin{aligned} h(ru_o) &= c_{i_o} \Phi_{x_{i_o}}(ru_o) \left(1 + \sum_{i \neq i_o} c_{i_o}^{-1} c_i \Phi_{x_i}(ru_o) / \Phi_{x_{i_o}}(ru_o) \right) \\ &= c_{i_o} \Phi_{x_{i_o}}(ru_o) \left(1 + \sum_{i \neq i_o} c_{i_o}^{-1} c_i \exp(\|x_{i_o}\|^2 - \|x_i\|^2) \exp(2r(x_i - x_{i_o})^\top u_o) \right) \\ &= c_{i_o} \Phi_{x_{i_o}}(ru_o) (1 + o(1)) \quad \text{as } r \rightarrow \infty, \end{aligned}$$

because $(x_i - x_{i_o})^\top u_o < 0$ for $i \neq i_o$.

General recipe 3 (Sums and products). It is clear that a linear combination of positive definite kernels with strictly positive scalars is a positive definite kernel too. And if one of the former kernels is strictly positive definite, then the linear combination is strictly positive as well.

Here is a result about products of kernels:

Theorem 3.3. *If k_1 and k_2 are positive definite kernels on $\mathcal{X}$, then their pointwise product $k_1 k_2$ is also a positive definite kernel. If k_1 and k_2 are strictly positive definite, then $k_1 k_2$ is strictly positive definite too.*

Using the characterization of (strict) positive definiteness in terms of Gram matrices, this result is an immediate consequence of the following result from linear algebra:

Theorem 3.4 (Schur). *Let A and B be symmetric and positive semidefinite matrices in $\mathbb{R}^{d \times d}$. Then their Hadamard product $A \odot B = (A_{ij} B_{ij})_{i,j=1}^d$ is also symmetric and positive semidefinite. If A and B are positive definite, then $A \odot B$ is positive definite too.*

Proof. For a vector $x \in \mathbb{R}^d$,

$$\begin{aligned}
x^\top A \odot B x &= \sum_{i,j=1}^n x_i A_{ij} B_{ij} x_j \\
&= \sum_{i,j=1}^n (\text{diag}(x)A)_{ij} (B \text{diag}(x))_{ij} \\
&= \text{trace}((\text{diag}(x)A)^\top B \text{diag}(x)) \\
&= \text{trace}(A \text{diag}(x) B \text{diag}(x)) \\
&= \text{trace}(A^{1/2} A^{1/2} \text{diag}(x) B^{1/2} B^{1/2} \text{diag}(x)) \\
&= \text{trace}(A^{1/2} \text{diag}(x) B^{1/2} B^{1/2} \text{diag}(x) A^{1/2}) \\
&= \text{trace}((B^{1/2} \text{diag}(x) A^{1/2})^\top (B^{1/2} \text{diag}(x) A^{1/2})) \\
&= \sum_{i,j=1}^n (B^{1/2} \text{diag}(x) A^{1/2})_{ij}^2 \\
&\geq 0.
\end{aligned}$$

This shows that $A \odot B$ is positive semidefinite.

If A and B are positive definite, then $A^{1/2}$ and $B^{1/2}$ are positive definite too, and for $x \neq 0$, the matrix $B^{1/2} \text{diag}(x) A^{1/2}$ is nonzero. For instance, $B^{1/2} \text{diag}(x) A^{1/2} A^{-1/2} x = B^{1/2} y$ with $y := (x_i^2)_{i=1}^n \neq 0$. Hence, $A \odot B$ is positive definite as well. $\square$

Example 3.5 (The standard Laplace kernel). For $x, y \in \mathbb{R}^d$, let

$$k(x, y) := \exp(-\|x - y\|_1)$$

with $\|z\|_1 := \sum_{i=1}^d |z_i|$ for $z \in \mathbb{R}^d$. This defines a strictly positive definite kernel k on $\mathbb{R}^d$. This can be shown as follows: For each $\sigma \in (0, \infty)^d$,

$$k_\sigma(x, y) := \prod_{i=1}^d (1 - |x_i - y_i|/\sigma_i)$$

is a strictly positive definite kernel on $\mathbb{R}^d$. Just note that $k_\sigma(x, y) = k_o(\text{diag}(\sigma)^{-1}x, \text{diag}(\sigma)^{-1}y)$ with the d -variate triangular kernel k_o on $\mathbb{R}^d$. But then, for any measure Π on $(0, \infty)^d$ with $0 < \Pi((0, \infty)^d) < \infty$,

$$(x, y) \mapsto \int_{(0, \infty)^d} k_\sigma(x, y) \Pi(d\sigma)$$

defines a strictly positive definite kernel on $\mathbb{R}^d$. Specifically, if Π has density

$$w(\sigma) := \prod_{i=1}^d \sigma_i \exp(-\sigma_i)$$

with respect to Lebesgue measure on $(0, \infty)^d$, then by Fubini's theorem,

$$\begin{aligned}
\int_{(0, \infty)^d} k_\sigma(x, y) \Pi(d\sigma) &= \prod_{i=1}^d \int_0^\infty s e^{-s} (1 - |x_i - y_i|/s)^+ ds \\
&= \prod_{i=1}^d \int_{|x_i - y_i|}^\infty (s - |x_i - y_i|) e^{-s} ds \\
&= \prod_{i=1}^d \exp(-|x_i - y_i|) \int_0^\infty v e^{-v} dv \\
&= \prod_{i=1}^d \exp(-|x_i - y_i|) = k(x, y).
\end{aligned}$$

Example 3.6 (Exponentials of kernels). Let k_o be a positive definite kernel on $\mathcal{X}$. Then

$$k(x, y) := \exp(k_o(x, y))$$

defines a positive definite kernel on $\mathcal{X}$ too. Indeed, for all $m \in \mathbb{N}$, points $x_1, \dots, x_m \in \mathcal{X}$ and scalars $c_1, \dots, c_m \in \mathbb{R}$,

$$\begin{aligned}
\sum_{i,j=1}^m c_i c_j \exp(k_o(x_i, x_j)) &= \sum_{i,j=1}^m c_i c_j + \sum_{\ell=1}^\infty \frac{1}{\ell!} \sum_{i,j=1}^m c_i c_j k_o^\ell(x_i, x_j) \\
&= \left(\sum_{i=1}^m c_i \right)^2 + \sum_{\ell=1}^\infty \frac{1}{\ell!} \sum_{i,j=1}^m c_i c_j k_o^\ell(x_i, x_j) \\
&\geq 0,
\end{aligned}$$

because k_o^ℓ is a positive definite kernel for each integer $\ell \geq 1$.

An interesting special case is that $(\mathcal{X}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ is a real pre-Hilbert space, and $k_o(x, y) = \langle x, y \rangle$, leading to

$$k(x, y) = \exp(\langle x, y \rangle).$$

Since $(x, y) \mapsto \langle x, y \rangle$ is a positive definite kernel on $\mathcal{X}$ (corresponding to the trivial feature mapping $x \mapsto \Phi_x := x$), this kernel k is positive definite. It is even strictly positive definite, because for arbitrary $m \in \mathbb{N}$, pairwise different points $x_1, \dots, x_m \in \mathcal{X}$ and real numbers $c_1, \dots, c_m \neq 0$,

$$\sum_{i,j=1}^m c_i c_j \exp(\langle x_i, x_j \rangle) = \sum_{i,j=1}^m \tilde{c}_i \tilde{c}_j \exp(-\|x_i - x_j\|^2/2)$$

with $\tilde{c}_i := c_i \exp(\|x_i\|^2/2) \neq 0$, and the sum on the right hand side is strictly positive, because the Gaussian kernel $(x, y) \mapsto \exp(-\|x - y\|^2/2)$ is strictly positive definite.

4 Some properties of RKHSs

Throughout this section, let $\mathcal{H} = (\mathcal{H}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ be a RKHS of functions on a set $\mathcal{X}$ with kernel k . In what follows, we investigate some properties of $\mathcal{H}$ and $\mathcal{X}$.

The kernel pseudometric on $\mathcal{X}$. While the set $\mathcal{X}$ is arbitrary, the kernel k and its RKHS $\mathcal{H}$ induce a pseudometric d_k on $\mathcal{X}$ via

$$d_k(x, y) := \|k(x, \cdot) - k(y, \cdot)\| = \sqrt{k(x, x) + k(y, y) - 2k(x, y)}.$$

Any function $f \in \mathcal{H}$ is Lipschitz continuous with respect to d_k , namely,

$$|f(x) - f(y)| = |\langle f, k(x, \cdot) - k(y, \cdot) \rangle| \leq \|f\| d_k(x, y).$$

If the kernel k is strictly positive definite, then d_k is a metric on $\mathbb{X}$.

Separability of $\mathcal{H}$. The RKHS $\mathcal{H}$ is separable if and only if $(\mathcal{X}, d_k)$ is separable. On the one hand, if $\mathcal{H}$ is separable, then its subset $\{k(x, \cdot) : x \in \mathcal{X}\}$ is separable too, and this implies that $(\mathcal{X}, d_k)$ is separable. On the other hand, let $\mathcal{X}_o$ be a countable, dense subset of $\mathcal{X}$ with respect to d_k . If $\mathcal{H}_{oo}$ denotes the set of all linear combinations of functions $k(x_o, \cdot)$, $x_o \in \mathcal{X}_o$, with rational coefficients, then $\mathcal{H}_{oo}$ is a countable, dense subset of $\mathcal{H}_o = \text{span}(k(x, \cdot) : x \in \mathcal{X})$, and since $\mathcal{H}_o$ is dense in $\mathcal{H}$, the set $\mathcal{H}_{oo}$ is a countable, dense subset of $\mathcal{H}$.

Suppose that $(\mathcal{X}, d)$ is a separable metric space, and suppose that $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is continuous with respect to the corresponding product topology on $\mathcal{X} \times \mathcal{X}$. Then $(\mathcal{X}, d_k)$ is separable too, and this implies that $\mathcal{H}$ is separable.

Sums of RKHSs. For $\ell = 1, 2$, let $\mathcal{H}_\ell = (\mathcal{H}_\ell, \langle \cdot, \cdot \rangle_\ell, \|\cdot\|_\ell)$ be a RKHS with reproducing kernel k_ℓ . We have mentioned already that $k := k_1 + k_2$ defines a positive definite kernel as well. Is it true, that the corresponding RKHS is the sum $\mathcal{H} := \mathcal{H}_1 + \mathcal{H}_2 = \{f_1 + f_2 : f_1 \in \mathcal{H}_1, f_2 \in \mathcal{H}_2\}$? The answer is yes, but we take a little detour to verify this.

Note first that the cartesian product $\mathcal{H}_{12} := \mathcal{H}_1 \times \mathcal{H}_2$ becomes a Hilbert space by defining

$$\langle (f_1, f_2), (g_1, g_2) \rangle_{12} := \langle f_1, g_1 \rangle_1 + \langle f_2, g_2 \rangle_2,$$

and $\|\cdot\|_{12}$ denotes the corresponding norm. Then $\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2$ is the image of the surjective linear mapping

$$S : \mathcal{H}_{12} \rightarrow \mathcal{H}, \quad S(f_1, f_2) := f_1 + f_2.$$

We have not specified a topology on $\mathcal{H}$ yet, but let us look at the kernel of S : The set $\mathcal{V} := S^{-1}(0)$ may be written as

$$\mathcal{V} = \{(h, -h) : h \in \mathcal{H}_1 \cap \mathcal{H}_2\},$$

and this can be shown to be a closed linear subspace of $\mathcal{H}_{12}$. Consequently, if Π denotes the orthogonal projection of $\mathcal{H}_{12}$ onto $\mathcal{V}^\perp$, then for any $f = S(f_1, f_2)$ with $(f_1, f_2) \in \mathcal{H}_{12}$,

$$S^{-1}(f) = (f_1, f_2) + \mathcal{V} = \Pi(f_1, f_2) + \mathcal{V},$$

and

$$\Pi(f_1, f_2) = \arg \min_{(g_1, g_2) \in S^{-1}(f)} \|(g_1, g_2)\|_{12}^2.$$

Consequently, for each $f \in \mathcal{H}$, there exists a unique point $T(f) \in S^{-1}(f) \cap \mathcal{V}^\perp$ such that $f = S(T(f))$. Now we can define an inner product on $\mathcal{H}$ as follows: For $f, g \in \mathcal{H}$,

$$\langle f, g \rangle := \langle T(f), T(g) \rangle_{12}.$$

This makes $\mathcal{H}$ a Hilbert space, and the corresponding norm $\| \cdot \|$ may be written as

$$\|f\|^2 = \min_{(f_1, f_2) \in \mathcal{H}_{12}: f_1 + f_2 \equiv f} (\|f_1\|_1^2 + \|f_2\|_2^2).$$

For any $x \in \mathcal{X}$ and $(f_1, f_2) \in \mathcal{H}_{12}$,

$$\begin{aligned} \langle (k_1(x, \cdot), k_2(x, \cdot)), (f_1, f_2) \rangle_{12} &= \langle k_1(x, \cdot), f_1 \rangle_1 + \langle k_2(x, \cdot), f_2 \rangle_2 \\ &= f_1(x) + f_2(x) \\ &= S(f_1, f_2)(x). \end{aligned}$$

In particular, for $(h, -h) \in \mathcal{V}$,

$$\langle (k_1(x, \cdot), k_2(x, \cdot)), (h, -h) \rangle_{12} = 0,$$

so $k = k_1 + k_2$ satisfies $T(k(x, \cdot)) = (k(x_1, \cdot), k_2(x, \cdot))$, and for any $f \in \mathcal{H}$,

$$\langle k(x, \cdot), f \rangle = f(x).$$

Characterizing functions in $\mathcal{H}$. Unfortunately, it is not easy to check whether a given function $f : \mathcal{X} \rightarrow \mathbb{R}$ belongs to $\mathcal{H}$ or not. A necessary condition is that for some real constant $C > 0$,

$$|f(x)| \leq C k(x, x)^{1/2} \quad \text{and} \quad |f(x) - f(y)| \leq C d_k(x, y) \quad \text{for all } x, y \in \mathcal{X}.$$

Here is a complete answer in terms of tensor products of functions. For arbitrary functions $f : \mathcal{X} \rightarrow \mathbb{R}$ and $g : \mathcal{Y} \rightarrow \mathbb{R}$, the product $f \otimes g$ denotes the function $\mathcal{X} \times \mathcal{Y} \ni (x, y) \mapsto f(x)g(y)$.

Theorem 4.1 (Krein). *A function $f : \mathbb{X} \rightarrow \mathbb{R}$ belongs to $\mathcal{H}$ if and only if for some $\epsilon > 0$,*

$$k - \epsilon f \otimes f$$

is a positive definite kernel on $\mathcal{X}$. In that case, $\epsilon \leq \|f\|^{-2}$.

Proof of Theorem 4.1. Suppose that $f : \mathcal{X} \rightarrow \mathbb{R}$ is an arbitrary function such that for some $\epsilon > 0$, the kernel $k_1 := k - \epsilon f \otimes f$ is positive definite. Then $k = k_1 + k_2$ with the kernel $k_2 := \epsilon f \otimes f$. The latter is also positive definite, because for any $m \in \mathbb{N}$, $x_1, \dots, x_m \in \mathcal{X}$ and $c_1, \dots, c_m \in \mathbb{R}$,

$$\sum_{i,j=1}^m c_i c_j k_2(x_i, x_j) = \epsilon \left(\sum_{i=1}^m c_i f(x_i) \right)^2 \geq 0.$$

Consequently, if $\mathcal{H}_1$ and $\mathcal{H}_2$ denote the RKHSs with kernels k_1 and k_2 , respectively, then $\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 \supset \mathcal{H}_2$, and $\mathcal{H}_2 = \text{span}(k_2(x, \cdot) : x \in \mathcal{X}) = \text{span}(\epsilon f(x)f : x \in \mathcal{X}) \ni f$.

Recall that $k - \epsilon f \otimes f$ is positive definite if and only if for any $m \in \mathbb{N}$, $x_1, \dots, x_m \in \mathcal{X}$ and $c_1, \dots, c_m \in \mathbb{R}$,

$$\sum_{i,j=1}^m c_i c_j (k(x_i, x_j) - \epsilon f(x_i) f(x_j)) = \sum_{i,j=1}^m c_i c_j k(x_i, x_j) - \epsilon \left(\sum_{i=1}^m c_i f(x_i) \right)^2 \geq 0.$$

If $f \in \mathcal{H}$, and if we define $g := \sum_{i=1}^m c_i k(x_i, \cdot)$, then the latter inequality may be rewritten as

$$\|g\|^2 - \epsilon \langle g, f \rangle^2 \geq 0.$$

Thus, for $f \in \mathcal{H}$, the kernel $k - \epsilon f \otimes f$ is positive definite if and only if

$$\epsilon \langle f, g \rangle^2 \leq \|g\|^2 \quad \text{for all } g \in \mathcal{H}_o.$$

Since $\mathcal{H}_o$ is dense in $\mathcal{H}$, the latter condition is equivalent to

$$\epsilon \langle f, g \rangle^2 \leq \|g\|^2 \quad \text{for all } g \in \mathcal{H},$$

and by means of the Cauchy–Schwarz inequality, this is equivalent to $\|f\|^2 \leq \epsilon^{-1}$. $\square$

5 Kernel mean embeddings

5.1 Definition of the embedding

Let $\mathcal{H} = (\mathcal{H}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ be a RKHS of functions on a set $\mathcal{X}$ with kernel k . Let us look at the dense linear subspace $\mathcal{H}_o = \text{span}(k(x, \cdot) : x \in \mathcal{X})$ once more. A function $h = \sum_{i=1}^m c_i k(x_i, \cdot)$ in $\mathcal{H}_o$ may be interpreted as

$$f(y) = \int k(x, y) \mathbb{S}(dx) \tag{5.1}$$

with the discrete signed measure

$$\mathbb{S} := \sum_{i=1}^m c_i \delta_{x_i}.$$

Here δ_x denotes Dirac measure at the point $x \in \mathcal{X}$, that is,

$$\begin{aligned} \mathbb{S}(B) &= \sum_{i=1}^m c_i 1_{[x_i \in B]} \quad \text{for } B \subset \mathcal{X}, \\ \int h(x) \mathbb{S}(dx) &= \sum_{i=1}^m c_i h(x_i) \quad \text{for } h : \mathcal{X} \rightarrow \mathbb{R}. \end{aligned}$$

An obvious question is whether we could consider more general signed measures $\mathbb{S}$ and define functions $f \in \mathcal{H}$ via (5.1). From now on, let $\mathcal{X}$ be equipped with a σ -field $\mathcal{B}$, and suppose that $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is measurable, where $\mathbb{X} \times \mathbb{X}$ is equipped with product σ -field $\mathcal{B} \otimes \mathcal{B}$. This implies that all functions in $\mathcal{H}_o$ are measurable, and since pointwise limits of sequences of measurable functions are measurable too, all functions in $\mathcal{H}$ are $\mathcal{B}$ -measurable.

Now consider first a σ -finite measure $\mathbb{P}$ on $\mathcal{X}$. The question is, whether $\int k(x, y) \mathbb{P}(dx)$ is well-defined for all $y \in \mathcal{X}$, and whether it is a function in $\mathcal{H}$. Since $|k(x, y)| \leq \sqrt{k(x, x)k(y, y)}$, a sufficient condition for $k(\cdot, y) \in L^1(\mathbb{P})$, $y \in \mathcal{X}$, is given by

$$\kappa_{\mathbb{P}} := \int \sqrt{k(x, x)} \mathbb{P}(dx) < \infty.$$

Then the integral $\int k(x, y) \mathbb{P}(dx)$ is well-defined in $\mathbb{R}$ with modulus no larger than $\kappa_{\mathbb{P}} \sqrt{k(y, y)}$. More generally, for any function $f \in \mathcal{H}$,

$$\int \langle f, k(x, \cdot) \rangle \mathbb{P}(dx)$$

is well-defined in $\mathbb{R}$, because

$$\int |\langle f, k(x, \cdot) \rangle| \mathbb{P}(dx) \leq \|f\| \int \|k(x, \cdot)\| \mathbb{P}(dx) = \|f\| \kappa_{\mathbb{P}}.$$

Moreover, this inequality shows that $f \mapsto \int \langle f, k(x, \cdot) \rangle \mathbb{P}(dx)$ defines a continuous linear form on $\mathcal{H}$. By Riesz' representation theorem, there exists a unique function $\mu_{\mathbb{P}} \in \mathcal{H}$ such that

$$\int \langle f, k(x, \cdot) \rangle \mathbb{P}(dx) = \langle f, \mu_{\mathbb{P}} \rangle \quad \text{for all } f \in \mathcal{H}.$$

In other words,

$$\mu_{\mathbb{P}} := \int k(x, \cdot) \mathbb{P}(dx),$$

where the integral on the right hand side is a Pettis integral, and

$$\|\mu_{\mathbb{P}}\| \leq \kappa_{\mathbb{P}}.$$

By plugging in $f = k(y, \cdot)$, we obtain the explicit representation

$$\mu_{\mathbb{P}}(y) = \int k(x, y) \mathbb{P}(dx) \quad \text{for all } y \in \mathcal{X}. \quad (5.2)$$

This definition can be extended to σ -finite signed measures $\mathbb{S}$. That means, for a σ -finite measure $\mathbb{P}$ and a measurable function $S : \mathcal{X} \rightarrow \mathbb{R}$ we define $\mathbb{S}(B) := \int_B S d\mathbb{P}$ and $\int h(x) \mathbb{S}(dx) := \int h(x) S(x) \mathbb{P}(dx)$ whenever these integrals are well-defined in $[-\infty, \infty]$. If this signed measure $\mathbb{S}$ satisfies the inequality

$$\kappa_{\mathbb{S}} := \int \sqrt{k(x, x)} |S(x)| \mathbb{P}(dx) < \infty,$$

then

$$\mu_{\mathbb{S}} := \int k(x, \cdot) \mathbb{S}(dx)$$

is the unique function in $\mathcal{H}$ such that

$$\langle f, \mu_{\mathbb{S}} \rangle = \int \langle f, k(x, \cdot) \rangle \mathbb{S}(dx) \quad \text{for all } f \in \mathcal{H}.$$

This function $\mu_{\mathbb{S}}$ satisfies the inequality

$$\|\mu_{\mathbb{S}}\| \leq \kappa_{\mathbb{S}},$$

and for any $y \in \mathcal{X}$,

$$\mu_{\mathbb{S}}(y) = \int k(x, y) \mathbb{S}(dx).$$

We call $\mu_{\mathbb{S}}$ the *kernel mean embedding* of $\mathbb{S}$.

The precise value of $\|\mu_{\mathbb{S}}\|$ is given by

$$\begin{aligned} \|\mu_{\mathbb{S}}\|^2 &= \langle \mu_{\mathbb{S}}, \mu_{\mathbb{S}} \rangle = \int \langle \mu_{\mathbb{S}}, k(x, \cdot) \rangle \mathbb{S}(dx) \\ &= \int \mu_{\mathbb{S}}(x) \mathbb{S}(dx) \\ &= \int \int k(x, y) \mathbb{S}(dy) \mathbb{S}(dx). \end{aligned}$$

Consequently, whenever $\mathbb{S}$ is a σ -finite signed measure such that $\kappa_{\mathbb{S}} < \infty$,

$$\kappa_{\mathbb{S}}^2 \geq \|\mu_{\mathbb{S}}\|^2 = \int \int k(x, y) \mathbb{S}(dx) \mathbb{S}(dy) \geq 0.$$

With the representation $\mathbb{S}(dx) = S(x) \mathbb{P}(dx)$ one can also write

$$\|\mu_{\mathbb{S}}\|^2 = \int \int k(x, y) S(x) S(y) \mathbb{P}(dx) \mathbb{P}(dy).$$

5.2 Integrally strictly positive definite kernels

From now on we consider a bounded positive definite and measurable kernel k on $\mathcal{X}$. Then $\mu_{\mathbb{S}} \in \mathcal{H}$ is well-defined for any finite signed measure $\mathbb{S}$ on $\mathcal{X}$. The space of finite signed measures on $\mathcal{X}$ is a real vector space on which $\mathbb{S} \mapsto \mu_{\mathbb{S}}$ acts linearly. This mapping is injective if and only if

$$\mu_{\mathbb{S}} \equiv 0 \quad \text{implies that} \quad \mathbb{S} \equiv 0.$$

Definition 5.1. A bounded positive definite kernel k on $\mathcal{X}$ is called *integrally strictly positive definite* if the mapping $\mathbb{S} \mapsto \mu_{\mathbb{S}}$ is injective on the space of all bounded signed measures on $\mathcal{X}$.

It is called *characteristic*, if the mapping $\mathbb{P} \mapsto \mu_{\mathbb{P}}$ is injective on the set of all probability distributions on $\mathcal{X}$.

Obviously, an integrally strictly positive definite kernel k is also characteristic. Moreover, it is strictly positive definite, because for $m \in \mathbb{N}$, pairwise different points $x_1, \dots, x_m \in \mathcal{X}$ and real numbers $c_1, \dots, c_m \neq 0$,

$$\sum_{i,j=1}^m c_i c_j k(x_i, x_j) = \|\mu_{\mathbb{S}}\|^2$$

with $\mathbb{S} = \sum_{i=1}^m c_i \delta_{x_i} \neq 0$.

5.3 Translation-invariant kernels on $\mathbb{R}^d$

Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a symmetric and *translation-invariant* function. That means,

$$k(x, y) = k(x + z, y + z) \quad \text{for arbitrary } x, y, z \in \mathbb{R}^d.$$

This is equivalent to saying that

$$k(x, y) = k_o(y - x)$$

for some even function $k_o : \mathbb{R}^d \rightarrow \mathbb{R}$. In fact, $k_o(z) = k(0, z) = k(z, 0)$ for $z \in \mathbb{R}^d$.

Suppose that k is a positive definite kernel. Then k_o is bounded, because for arbitrary $x \in \mathbb{R}^d$,

$$k_o(x)^2 = k(0, x)^2 \leq k(0, 0)k(x, x) = k_o(0)^2.$$

Moreover, if k_o is continuous at 0, then it is uniformly continuous on $\mathbb{R}^d$. Indeed, if $(\mathcal{H}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ is the RKHS of k , then for $x, y \in \mathbb{R}^d$,

$$\begin{aligned} |k_o(x) - k_o(y)| &= |k(0, x) - k(0, y)| \\ &= |\langle k(0, \cdot), k(x, \cdot) - k(y, \cdot) \rangle| \\ &\leq \|k(0, \cdot)\| \|k(x, \cdot) - k(y, \cdot)\| \\ &= \sqrt{2k_o(0)(k_o(0) - k_o(x - y))}, \end{aligned}$$

and the right hand side converges to 0 as $x - y \rightarrow 0$.

Bochner's theorem. A first natural question is whether a given even function $k_o : \mathbb{R}^d \rightarrow \mathbb{R}$ gives rise to a positive definite kernel k via $k(x, y) := k_o(y - x)$. This question has a complete answer if k_o is continuous.

Theorem 5.2 (Bochner). *Let $k_o : \mathbb{R}^d \rightarrow \mathbb{R}$ be an even, continuous function. Then $k(x, y) := k_o(x - y)$ defines a positive definite kernel if and only if there exists a finite, symmetric measure Λ on $\mathbb{R}^d$ such that k_o is the characteristic function of Λ ,*

$$k_o(x) = \int_{\mathbb{R}^d} \exp(\mathbf{i} t^\top x) \Lambda(dt)$$

for all $x \in \mathbb{R}^d$.

Symmetry of Λ means that $\Lambda(-B) = \Lambda(B)$ for any Borel subset B of $\mathbb{R}^d$. In particular,

$$\int_{\mathbb{R}^d} \exp(\mathbf{i} t^\top x) \Lambda(dt) = \int_{\mathbb{R}^d} \cos(t^\top x) \Lambda(dt).$$

A sufficient condition for strict positive definiteness. A second natural question is whether a translation-invariant positive definite kernel k is integrally strictly positive definite. If $k_o := k(0, \cdot)$ is integrable, there is a rather general criterion for this desirable property in terms of the characteristic function (or Fourier transform) of k_o .

Theorem 5.3. Suppose that $k_o \in \mathcal{L}^1(\mathbb{R}^d)$ yields a translation-invariant positive definite kernel k . Let $\phi_o : \mathbb{R}^d \rightarrow \mathbb{C}$ be the characteristic function of k_o ,

$$\phi_o(t) := \int_{\mathbb{R}^d} \exp(\mathbf{i} x^\top t) k_o(x) dx \in \mathbb{C}.$$

If $\{t \in \mathbb{R}^d : \phi_o(t) \neq 0\}$ is a dense subset of $\mathbb{R}^d$, then $k(x, y)$ is integrally strictly positive definite.

Fortunately, one does not have to compute the characteristic function ϕ_o explicitly to apply this theorem.

Corollary 5.4. Suppose that $k_o \in \mathcal{L}^1(\mathbb{R}^d)$ yields a translation-invariant positive definite kernel k . If $\text{Leb}(\{k_o \neq 0\}) > 0$ and $\int \exp(\epsilon \|x\|) |k_o(x)| dx < \infty$ for some $\epsilon > 0$, then k is integrally strictly positive definite.

By means of this corollary, one sees immediately that the triangular kernel, the Laplace kernel or the standard Gaussian kernel on $\mathbb{R}^d$ are integrally strictly positive definite.

Proof of Theorem 5.3. Note that a finite signed measure $\mathbb{S}$ on $\mathbb{R}^d$ is uniquely determined by its characteristic function $\phi_{\mathbb{S}} : \mathbb{R}^d \rightarrow \mathbb{C}$ with

$$\phi_{\mathbb{S}}(t) = \int_{\mathbb{R}^d} \exp(\mathbf{i} x^\top t) \mathbb{S}(dx).$$

Moreover, ϕ_o is the characteristic function of the finite signed measure K_o with density k_o with respect to Lebesgue measure, and $\mu_{\mathbb{S}}$ is the density of the convolution $\mathbb{S} \star K_o$. To see this, note that for any measurable and bounded function $h : \mathbb{R}^d \rightarrow \mathbb{R}$, the definitions of $\mathbb{S} \star K_o$ and $\mu_{\mathbb{S}}$ imply that

$$\begin{aligned} \int_{\mathbb{R}^d} h(x) \mathbb{S} \star K_o(dx) &:= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(x_1 + x_2) \mathbb{S}(dx_1) K_o(dx_2) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(x_1 + x_2) k_o(x_2) dx_2 \mathbb{S}(dx_1) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(y) k_o(y - x_1) dy \mathbb{S}(dx_1) \\ &= \int_{\mathbb{R}^d} h(y) \int_{\mathbb{R}^d} h(y) k_o(y - x_1) \mathbb{S}(dx_1) dy \\ &= \int_{\mathbb{R}^d} h(y) \mu_{\mathbb{S}}(y) dy. \end{aligned}$$

Consequently, $\mu_{\mathbb{S}} \equiv 0$ is equivalent to $\mathbb{S} \star K_o \equiv 0$. The characteristic function of $\mathbb{S} \star K_o$ is given by the pointwise product of the characteristic functions of $\mathbb{S}$ and K_o ,

$$\phi_{\mathbb{S} \star K_o}(t) = \phi_{\mathbb{S}}(t) \phi_o(t)$$

for all $t \in \mathbb{R}^d$. Indeed,

$$\begin{aligned}
\phi_{\mathbb{S} \star K_o}(t) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \exp(\mathbf{i}(x_1 + x_2)^\top t) \mathbb{S}(dx_1) K_o(dx_2) \\
&= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \exp(\mathbf{i}x_1^\top t) \exp(\mathbf{i}x_2^\top t) \mathbb{S}(dx_1) K_o(dx_2) \\
&= \int_{\mathbb{R}^d} \exp(\mathbf{i}x_1^\top t) \mathbb{S}(dx_1) \int_{\mathbb{R}^d} \exp(\mathbf{i}x_2^\top t) K_o(dx_2) \\
&= \phi_{\mathbb{S}}(t) \phi_o(t).
\end{aligned}$$

Since $\phi_{\mathbb{S}}$ is continuous, and since $\phi_o \neq 0$ on a dense subset of $\mathbb{R}^d$, it follows from $\phi_{\mathbb{S} \star K_o} \equiv 0$ that $\phi_{\mathbb{S}} \equiv 0$. But this is equivalent to $\mathbb{S} \equiv 0$. $\square$

Proof of Corollary 5.4. Note that $\phi_o(t) = \int_{\mathbb{R}^d} \exp(\mathbf{i}t^\top x) k_o(x) dx$ is well-defined for all complex vectors $t \in V^d$, where $V := \{z \in \mathbb{C} : |\operatorname{Im} z| < \epsilon_o\}$, where $\epsilon_o = d^{-1/2}\epsilon$. Moreover, for any index $i_o \in \{1, \dots, d\}$, $\phi_o(t)$ is holomorphic in $t_{i_o} \in V$ if the other components $t_i \in V$ are fixed. Recall that a holomorphic function f on V is constant zero if, say, $f = 0$ on a nondegenerate real interval. Now suppose that the set $\{t \in \mathbb{R}^d : \phi_o(t) \neq 0\}$ is *not* dense in $\mathbb{R}^d$. This is equivalent to saying that $\phi_o(t) = 0$ for all t in $U_1 \times U_2 \times \dots \times U_d \subset \mathbb{R}^d$ with nondegenerate intervals $U_i \subset \mathbb{R}$. By fixing all but the first component of an arbitrary $t \in U$, one can deduce that U_1 may be replaced with $\mathbb{R}$. Then fixing all but the second component of an arbitrary $t \in U$, one may replace U_2 with $\mathbb{R}$, and so on. Eventually we know that $\phi_o \equiv 0$. But this would be equivalent to $k_o = 0$ almost everywhere w.r.t. Lebesgue measure, a contradiction to $\operatorname{Leb}(\{k_o \neq 0\}) > 0$. $\square$

Example 5.5 (A variation of the standard Gaussian kernel on $\mathbb{R}^d$). We have seen already that the standard Gaussian kernel k on $\mathbb{R}^d$, $k(x, y) := \exp(-\|x - y\|^2/2)$ with the standard Euclidean norm $\|\cdot\|$ on $\mathbb{R}^d$, may be written as $k(x, y) = \langle \Phi_x, \Phi_y \rangle$ with the feature vectors $\Phi_x, \Phi_y \in (\mathcal{L}^2 \cap \mathcal{C})(\mathbb{R}^d)$ given by $\Phi_x(t) := D \exp(-\|t - x\|^2)$, $D > 0$ a suitable constant. Let us repeat this game with a different feature mapping. Let $(\mathcal{L}^2 \cap \mathcal{C})(\mathbb{R}^d, \mathbb{R}^d)$ be the space of all continuous functions $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $\int_{\mathbb{R}^d} \|g(t)\|^2 dt < \infty$. Then $(\mathcal{L}^2 \cap \mathcal{C})(\mathbb{R}^d, \mathbb{R}^d)$ becomes a pre-Hilbert space via $\langle g, h \rangle := \int_{\mathbb{R}^d} g(t)^\top h(t) dt$. Specifically, for $x \in \mathbb{R}^d$, let $\Phi_x \in (\mathcal{L}^2 \cap \mathcal{C})(\mathbb{R}^d, \mathbb{R}^d)$ be given by

$$\Phi_x(t) := D(t - x) \exp(-\|t - x\|^2)$$

for some constant $D > 0$ to be specified later. Then for $x, y \in \mathbb{R}^d$ and $m := 2^{-1}(x + y)$,

$$\delta := 2^{-1}(x - y),$$

$$\begin{aligned} \langle \Phi_x, \Phi_y \rangle &= D^2 \int_{\mathbb{R}^d} (t - x)^\top (t - y) \exp(-\|t - x\|^2 - \|t - y\|^2) dt \\ &= D^2 \int_{\mathbb{R}^d} (\|t - m\|^2 - \|\delta\|^2) \exp(-2\|t - m\|^2 - 2\|\delta\|^2) dt \\ &= (D/2)^2 \exp(-\|x - y\|^2/2) \int_{\mathbb{R}^d} (4\|t - m\|^2 - \|x - y\|^2) \exp(-2\|t - m\|^2) dt \\ &= (D/2)^2 2^{-d} \exp(-\|x - y\|^2/2) \int_{\mathbb{R}^d} (\|z\|^2 - \|x - y\|^2) \exp(-\|z\|^2/2) dz \\ &= (D/2)^2 (\pi/2)^{d/2} (d - \|x - y\|^2) \exp(-\|x - y\|^2/2). \end{aligned}$$

Taking $D := 2(2/\pi)^{d/4}$ shows that

$$k(x, y) := (d - \|x - y\|^2) \exp(-\|x - y\|^2/2)$$

defines a positive definite, translation-invariant kernel on $\mathbb{R}^d$. Corollary 5.4 reveals that k is even integrally strictly positive definite.

5.4 The standard Gaussian kernel on a Hilbert space

Let $\mathcal{X} = (\mathcal{X}, \langle \cdot, \cdot \rangle_{\mathcal{X}}, \|\cdot\|_{\mathcal{X}})$ be a separable real Hilbert space, and let k be the standard Gaussian kernel on $\mathcal{X}$,

$$k(x, y) := \exp(-\|x - y\|_{\mathcal{X}}^2/2).$$

Theorem 5.6. *The standard Gaussian kernel k on $\mathcal{X}$ is integrally strictly positive definite.*

We know already that this theorem is true in case of $\dim(\mathcal{X}) < \infty$. In the infinite-dimensional setting, it will be derived from the following generalization of a well-known result from probability theory about moment-generating functions (or Laplace transforms).

Proposition 5.7. *Let $\mathbb{Q}_1$ and $\mathbb{Q}_2$ be finite measures on $\mathcal{X}$, and let*

$$m_j(y) := \int \exp(\langle x, y \rangle_{\mathcal{X}}) \mathbb{Q}_j \in [0, \infty]$$

for $j = 1, 2$ and $y \in \mathcal{X}$. Suppose that for some $\epsilon > 0$, $m_1(y) = m_2(y) < \infty$ whenever $\|y\| \leq \epsilon$. Then $\mathbb{Q}_1 \equiv \mathbb{Q}_2$.

Proof of Theorem 5.6. Let $\mathbb{S} = \mathbb{Q}_1 - \mathbb{Q}_2$ with finite measures $\mathbb{Q}_1, \mathbb{Q}_2$ on $\mathcal{X}$. (If $\mathbb{S}(dx) = S(x) M(dx)$ for some σ -finite measure M on $\mathcal{X}$ and a function $S \in L^1(M)$, one could take $\mathbb{Q}_1(dx) := S^+(x) M(dx)$ and $\mathbb{Q}_2(dx) := S^-(x) M(dx)$.) Then

$$\mu_{\mathbb{S}}(y) = \exp(-\|y\|_2^2/2) \left(\int \exp(\langle x, y \rangle_{\mathcal{X}}) \tilde{\mathbb{Q}}_1(dx) - \int \exp(\langle x, y \rangle_{\mathcal{X}}) \tilde{\mathbb{Q}}_2(dx) \right)$$

with the finite signed measures $\tilde{\mathbb{Q}}_1, \tilde{\mathbb{Q}}_2$ given by $\tilde{\mathbb{Q}}_j(dx) := \exp(-\|x\|_2^2/2) \mathbb{Q}_j(dx)$. Thus, $\mu_{\mathbb{S}} \equiv 0$ is equivalent to

$$\int \exp(\langle x, y \rangle_{\mathcal{X}}) \tilde{\mathbb{Q}}_1(dx) = \int \exp(\langle x, y \rangle_{\mathcal{X}}) \tilde{\mathbb{Q}}_2(dx)$$

for arbitrary $y \in \mathcal{X}$. Note that by definition of the measures $\tilde{\mathbb{Q}}_j$, the latter integrals are finite. It follows from Proposition 5.7 that $\tilde{\mathbb{Q}}_1 \equiv \tilde{\mathbb{Q}}_2$. But $\mathbb{Q}_j$ may be represented as $\mathbb{Q}_j(dx) = \exp(\|x\|_{\mathcal{X}}^2/2) \tilde{\mathbb{Q}}_j(dx)$, so $\mathbb{Q}_1 \equiv \mathbb{Q}_2$, that is, $\mathbb{S} \equiv 0$. $\square$

Proof of Proposition 5.7. Note that $m_j(0) = \mathbb{Q}_j(\mathcal{X})$, so $\mathbb{Q}_1(\mathcal{X}) = \mathbb{Q}_2(\mathcal{X})$, and we may assume without loss of generality that $\mathbb{Q}_1, \mathbb{Q}_2$ are probability measures on $\mathcal{X}$.

If $\dim(\mathcal{X}) = d < \infty$, we may identify $\mathcal{X}$ with $\mathbb{R}^d$, equipped with its standard inner product. In that case, the result is well-known from probability theory: $m_j(y) = \phi_j(-iy)$, where ϕ_j is the characteristic function of $\mathbb{Q}_j$. Similarly as in the proof of Corollary 5.4, one can deduce from the equality of m_1 and m_2 on an open neighborhood of 0 in $\mathbb{R}^d$ that $\phi_1 \equiv \phi_2$ on $\mathbb{R}^d$, and this implies that $\mathbb{Q}_1 \equiv \mathbb{Q}_2$.

Let $\dim(\mathbb{X}) = \infty$. Separability of $\mathcal{X}$ implies the existence of a complete orthonormal system $(\phi_i)_{i \in \mathbb{N}}$ in $\mathcal{X}$. For $n \in \mathbb{N}$, let Π_n be the orthogonal projection from $\mathcal{X}$ onto the n -dimensional linear subspace $\mathcal{X}_n := \text{span}(\phi_1, \dots, \phi_n)$. If X_1 and X_2 are random variables with distributions $\mathbb{Q}_1$ and $\mathbb{Q}_2$, respectively, then

$$m_j(y) = \mathbb{E} \exp(\langle X_j, y \rangle_{\mathcal{X}})$$

for $j = 1, 2$ and any $y \in \mathcal{X}$. Specifically, if $y \in \mathcal{X}_n$ for some $n \in \mathbb{N}$, then

$$m_j(y) = \mathbb{E} \exp(\langle \Pi_n X_j, y \rangle_{\mathcal{X}}).$$

Consequently, if $m_1(y) = m_2(y) < \infty$ whenever $\|y\| \leq \epsilon$, then $\Pi_n X_1$ and $\Pi_n X_2$ have the same distribution for any $n \in \mathbb{N}$. But this implies that $\mathbb{Q}_1 \equiv \mathbb{Q}_2$, because $\Pi_n X_j \rightarrow X_j$ almost surely as $n \rightarrow \infty$ for $j = 1, 2$. Hence, for any bounded continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$, by dominated convergence,

$$\int f(x) \mathbb{Q}_1(dx) = \lim_{n \rightarrow \infty} \mathbb{E} f(\Pi_n X_1) = \lim_{n \rightarrow \infty} \mathbb{E} f(\Pi_n X_2) = \int f(x) \mathbb{Q}_2(dx).$$

$\square$

5.5 Application to the comparison of two samples

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random variables, where $X_i \sim \mathbb{P}$ and $Y_j \sim \mathbb{Q}$ with unknown distributions $\mathbb{P}$ and $\mathbb{Q}$ on a measurable space $\mathcal{X} = (\mathcal{X}, \mathcal{B})$. A classical testing problem is to detect whether $\mathbb{P} \neq \mathbb{Q}$. Suppose we have a bounded, measurable and characteristic kernel k on $\mathcal{X}$ at hand. Since $\mathbb{P} \neq \mathbb{Q}$ if and only if $\mu_{\mathbb{P}} \neq \mu_{\mathbb{Q}}$, a natural test statistic might be

$$T = T_{m,n}(X_1, \dots, X_m, Y_1, \dots, Y_n) := \|\mu_{\hat{\mathbb{P}}} - \mu_{\hat{\mathbb{Q}}}\|^2$$

with the empirical distributions $\hat{\mathbb{P}} := m^{-1} \sum_{i=1}^m \delta_{X_i}$ and $\hat{\mathbb{Q}} := n^{-1} \sum_{j=1}^n \delta_{Y_j}$ of the two samples. Consistency properties of $\mu_{\hat{\mathbb{P}}}$ and $\mu_{\hat{\mathbb{Q}}}$ as point estimators of $\mu_{\mathbb{P}}$ and $\mu_{\mathbb{Q}}$, respectively, will be treated

in Section C. But let us talk about computational and practical aspects of the two-sample test now. Applying the explicit formula for $\|\mu_{\mathbb{S}}\|^2$ with $\mathbb{S} = \widehat{\mathbb{P}} - \widehat{\mathbb{Q}}$ leads to the expression

$$T = \frac{1}{m^2} \sum_{i(1), i(2)=1}^m k(X_{i(1)}, X_{i(2)}) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(X_i, Y_j) + \frac{1}{n^2} \sum_{j(1), j(2)=1}^n k(Y_{j(1)}, Y_{j(2)}).$$

By means of the the pooled sample

$$\mathbf{Z} = (Z_1, \dots, Z_N) := (X_1, \dots, X_m, Y_1, \dots, Y_n)$$

of all $N := m + n$ observations, one can also write

$$\begin{aligned} T = T(\mathbf{Z}) &= \sum_{a,b=1}^N k(Z_a, Z_b) \left(\frac{1_{[a \leq m]}}{m} - \frac{1_{[a > m]}}{n} \right) \left(\frac{1_{[b \leq m]}}{m} - \frac{1_{[b > m]}}{n} \right) \\ &= \mathbf{v}^\top \mathbf{K} \mathbf{v} \end{aligned}$$

with the Gram matrix

$$\mathbf{K} := (k(Z_a, Z_b))_{a,b=1}^N \in \mathbb{R}^{N \times N}$$

and the auxiliary vector

$$\mathbf{v} := \left(\frac{1_{[a \leq m]}}{m} - \frac{1_{[a > m]}}{n} \right)_{a=1}^N \in \mathbb{R}^N.$$

This representation of T is particularly convenient to perform a Monte Carlo permutation test of our null hypothesis: The idea is to generate M independent random permutations $\mathbf{Z}_{[1]}, \mathbf{Z}_{[2]}, \dots, \mathbf{Z}_{[M]}$ of the given pooled data $\mathbf{Z}$, to compute the resulting values $T_s = T(\mathbf{Z}_{[s]})$ and to determine the Monte Carlo p-value

$$\widehat{\pi} := \frac{\#\{s \leq M : T_s \geq T\} + 1}{M + 1}.$$

This is an exact p-value in the sense that under the null hypothesis that $\mathbb{P} \equiv \mathbb{Q}$,

$$\text{prob}(\widehat{\pi} \leq \alpha) \leq \frac{\lfloor (M + 1)\alpha \rfloor}{M + 1} \leq \alpha,$$

where $\text{prob}(\cdot)$ takes into account the randomness in the data and in the simulations. Instead of permuting the pooled sample, however, it suffices to generate (simulate) M random permutations $\mathbf{v}_{[1]}, \mathbf{v}_{[2]}, \dots, \mathbf{v}_{[M]}$ of the vector $\mathbf{v}$ and to compute $T_s = \mathbf{v}_{[s]}^\top \mathbf{K} \mathbf{v}_{[s]}$.

Two examples. In the special case of $\mathcal{X} = \mathbb{R}^d$, one could think about test statistics involving kernel density estimators. For some bandwidth $\sigma > 0$, let γ_σ be the probability density of $\mathcal{N}_d(0, \sigma^2 I_d)$. Assuming that $\mathbb{P}$ and $\mathbb{Q}$ have unknown probability densities f and g , respectively, kernel density estimators of f and g are given by

$$\widehat{f}_\sigma(x) := \frac{1}{m} \sum_{i=1}^m \gamma_\sigma(x - X_i) \quad \text{and} \quad \widehat{g}_\sigma(x) := \frac{1}{n} \sum_{j=1}^n \gamma_\sigma(x - Y_j).$$

Possible test statistics for the null hypothesis that $\mathbb{P} \equiv \mathbb{Q}$ are given by

$$T^{(0)} := \int_{\mathbb{R}^d} |\widehat{f}_\sigma(x) - \widehat{g}_\sigma(x)|^2 dx \quad \text{and} \quad T^{(1)} := \int_{\mathbb{R}^d} \|\nabla \widehat{f}_\sigma(x) - \nabla \widehat{g}_\sigma(x)\|_2^2 dx$$

with the usual Euclidean norm $\|\cdot\|_2$ on $\mathbb{R}^d$. Fortunately, the explicit computation of these test statistics can be easily performed. Tedious but elementary calculations show that

$$\begin{aligned} k_\sigma^{(0)}(x, y) &:= \int_{\mathbb{R}^d} \gamma_\sigma(x-t) \gamma_\sigma(y-t) dt \\ &= (4\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|x-y\|_2^2}{4\sigma^2}\right), \\ k_\sigma^{(1)}(x, y) &:= \int_{\mathbb{R}^d} \nabla \gamma_\sigma(x-t)^\top \nabla \gamma_\sigma(y-t) dt \\ &= (4\pi\sigma^2)^{-d/2} (2\sigma^2)^{-1} \left(d - \frac{\|x-y\|_2^2}{2\sigma^2}\right) \exp\left(-\frac{\|x-y\|_2^2}{4\sigma^2}\right). \end{aligned}$$

Consequently,

$$T^{(j)} = \mathbf{v}^\top \mathbf{K}_\sigma^{(j)} \mathbf{v}$$

with $\mathbf{v} := (1_{[a \leq m]}/m - 1_{[a > m]}/n)_{a=1}^N$ and $\mathbf{K}_\sigma^{(j)} := (k_\sigma^{(j)}(Z_a, Z_b))_{a,b=1}^N$.

Note that $k_\sigma^{(j)}$ is a translation-invariant, integrally strictly positive definite kernel on $\mathbb{R}^d$, because it is just a rescaled version of the standard Gaussian kernel on $\mathbb{R}^d$ or its variation introduced as Example 5.5.

6 Mercer's theorem

Throughout this section let $(\mathcal{X}, d)$ be a separable metric space, and let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a bounded, continuous function and an integrally strictly positive definite kernel with corresponding RKHS $\mathcal{H} = (\mathcal{H}, \langle \cdot, \cdot \rangle, \|\cdot\|)$. Recall that all functions in $\mathcal{H}$ are bounded and Lipschitz continuous with respect to the kernel metric d_k on $\mathcal{X}$, and continuity of k implies that all functions in $\mathcal{H}$ are continuous with respect to the given metric d . Moreover, separability of $(\mathcal{X}, d)$ implies that $\mathcal{H}$ is separable, and for any finite measure $\mathbb{P}$ on (the Borel subsets of) $\mathcal{X}$, the space $L^2(\mathbb{P})$ is separable as well.

In this section we prove a variant of Mercer's theorem which applies in many relevant settings. Mercer proved it in the special case of a compact metric space and a probability measure $\mathbb{P}$ on $\mathcal{X}$ with full support.

Theorem 6.1 (Mercer). *Let $\mathbb{P}$ be a finite measure on $\mathcal{X}$. Let $\mathcal{X}_o$ be the support of $\mathbb{P}$, that is, the smallest closed set $\mathcal{X}_o \subset \mathcal{X}$ such that $\mathbb{P}(\mathcal{X} \setminus \mathcal{X}_o) = 0$. Then there exists a complete orthonormal system $(f_j)_{j \in J}$ in $L^2(\mathbb{P})$ and a tuple $(\lambda_j)_{j \in J} \in (0, \infty)^J$ with the following properties:*

$$\sum_{j \in J} \lambda_j = \bar{\kappa}_{\mathbb{P}} := \int k(x, x) \mathbb{P}(dx)$$

and

$$k = \sum_{j \in J} \lambda_j f_j \otimes f_j$$

in $L^2(\mathbb{P} \otimes \mathbb{P})$, where $\|k\|_{2, \mathbb{P} \otimes \mathbb{P}}^2 \leq \bar{\kappa}_{\mathbb{P}}^2$.

There exist unique versions $f_j \in \mathcal{H}$ of the $f_j \in L^2(\mathbb{P})$ such that

$$\int k(x, y) f_j(x) \mathbb{P}(dx) = \lambda_j f_j(y) \quad \text{for all } j \in J, y \in \mathcal{X},$$

and

$$k(x, y) = \sum_{j \in J} \lambda_j f_j(x) f_j(y) \quad \text{for all } x \in \mathcal{X}_o, y \in \mathcal{X}.$$

The functions $h_j := \lambda_j^{1/2} f_j$ form an orthonormal system in $\mathcal{H}$, and

$$\overline{\text{span}}(k(x, \cdot) : x \in \mathcal{X}_o) = \overline{\text{span}}(h_j : j \in J).$$

In particular, if $\mathcal{X}_o = \mathcal{X}$, that is, $\mathbb{P}(U) > 0$ for any nonempty open set $U \subset \mathcal{X}$, then

$$\mathcal{H} = \overline{\text{span}}(h_j : j \in J).$$

Proof of Theorem 6.1. Recall that $\mathcal{H} \subset \mathcal{L}^2(\mathbb{P})$. Precisely, for $h \in \mathcal{H}$ and $x \in \mathcal{X}$,

$$|h(x)| = |\langle h, k(x, \cdot) \rangle| \leq \|h\| \sqrt{k(x, x)},$$

whence $\|h\|_{2, \mathbb{P}}^2 \leq \bar{\kappa}_{\mathbb{P}} \|h\|^2$. In other words,

$$I_k h := [h] = \{g \in \mathcal{L}^2(\mathbb{P}) : \mathbb{P}(\{g \neq h\}) = 0\},$$

defines a continuous linear operator $I_k : \mathcal{H} \rightarrow L^2(\mathbb{P})$. It is even a Hilbert–Schmidt operator with

$$\|I_k\|_{\text{HS}}^2 = \bar{\kappa}_{\mathbb{P}}.$$

Indeed, for any complete orthonormal system $(h_a)_{a \in A}$ in $\mathcal{H}$,

$$\begin{aligned} \sum_{a \in A} \|I_k h_a\|_{2, \mathbb{P}}^2 &= \sum_{a \in A} \int h_a(x)^2 \mathbb{P}(dx) \\ &= \sum_{a \in A} \int \langle h_a, k(x, \cdot) \rangle^2 \mathbb{P}(dx) \\ &= \int \sum_{a \in A} \langle h_a, k(x, \cdot) \rangle^2 \mathbb{P}(dx) \\ &= \int \|k(x, \cdot)\|^2 \mathbb{P}(dx) \\ &= \int k(x, x) \mathbb{P}(dx) \\ &= \bar{\kappa}_{\mathbb{P}}. \end{aligned}$$

Now we look at the adjoint operator $S_k : L^2(\mathbb{P}) \rightarrow \mathcal{H}$ of I_k . It is uniquely determined by the equation

$$\langle h, S_k f \rangle = \langle I_k h, f \rangle_{\mathbb{P}} = \int h(x) f(x) \mathbb{P}(dx),$$

and with $h = k(y, \cdot)$ one sees that the function $S_k f \in \mathcal{H}$ is given by

$$S_k f(y) = \int k(x, y) f(x) \mathbb{P}(dx).$$

In other words, $S_k f$ is the kernel mean embedding $\mu_{f \cdot \mathbb{P}}$ of the finite signed measure $f \cdot \mathbb{P}$ with density $f \in L^2(\mathbb{P}) \subset L^1(\mathbb{P})$ with respect to $\mathbb{P}$. As shown in the proof of Theorem B.1, S_k is also a Hilbert–Schmidt operator with

$$\|S_k\|_{\text{HS}}^2 = \|I_k\|_{\text{HS}}^2 = \bar{\kappa}_{\mathbb{P}}.$$

Combining the two operators I_k and S_k leads to the integral operator $T_k := I_k S_k : L^2(\mathbb{P}) \rightarrow L^2(\mathbb{P})$, where $T_k f = [S_k f]$, and

$$\langle T_k f, g \rangle_{\mathbb{P}} = \int \int f(x) g(y) k(x, y) \mathbb{P}(dx) \mathbb{P}(dy) = \langle \mu_{f \cdot \mathbb{P}}, \mu_{g \cdot \mathbb{P}} \rangle.$$

This operator is compact, self-adjoint and strictly positive definite, see also Sections A and B, where strict positive definiteness follows from the assumption that k is integrally strictly positive definite. Consequently, there exists a complete orthonormal system $(f_j)_{j \in J}$ in $L^2(\mathbb{P})$ and a bounded tuple $(\lambda_j)_{j \in J} \in (0, \infty)^J$ such that

$$T_k = \sum_{j \in J} \lambda_j \langle \cdot, f_j \rangle_{\mathbb{P}} f_j,$$

and

$$k = \sum_{j \in J} \lambda_j f_j \otimes f_j$$

in $L^2(\mathbb{P} \otimes \mathbb{P})$. The eigenvalues $\lambda_j, j \in J$, satisfy even the condition

$$\sum_{j \in J} \lambda_j = \sum_{j \in J} \langle f_j, T_k f_j \rangle_{\mathbb{P}} = \sum_{j \in J} \langle f_j, I_k S_k f_j \rangle_{\mathbb{P}} = \sum_{j \in J} \langle S_k f_j, S_k f_j \rangle = \|S_k\|_{\text{HS}}^2 = \bar{\kappa}_{\mathbb{P}}.$$

That means, T_k satisfies an even stronger condition than being a Hilbert–Schmidt operator, and $\|k\|_{2, \mathbb{P}}^2 = \|T_k\|_{\text{HS}}^2 = \sum_{j \in J} \lambda_j^2$ is clearly not larger than $\bar{\kappa}_{\mathbb{P}}^2$.

An explicit representer of the equivalence class $f_j \in L^2(\mathbb{P})$ is given by $f_j(y) = \lambda_j^{-1} S_k f_j(y)$, a function in $\mathcal{H} \subset \mathcal{L}^2(\mathbb{P})$. Then we have the pointwise equation $\int k(x, y) f_j(x) \mathbb{P}(dx) = \lambda_j f_j(y)$ for all $j \in J$ and $y \in \mathcal{X}$. The functions $h_j := \lambda_j^{-1/2} S_k f_j, j \in J$, comprise an orthonormal system in $\mathcal{H}$, because for $i, j \in J$,

$$\langle h_i, h_j \rangle = \lambda_i^{-1/2} \lambda_j^{-1/2} \langle S_k f_i, S_k f_j \rangle = \lambda_i^{-1/2} \lambda_j^{-1/2} \langle T_k f_i, f_j \rangle_{\mathbb{P}} = 1_{[i=j]}.$$

The representation

$$\{k(x_o, \cdot) : x_o \in \mathcal{X}_o\}^\perp = \{h \in \mathcal{H} : h \equiv 0 \text{ on } \mathcal{X}_o\}$$

implies that

$$\overline{\text{span}}(h_j : j \in J) = \overline{\text{span}}(k(x_o, \cdot) : x_o \in \mathcal{X}_o).$$

To see this, note that for $h \in \mathcal{H}$, completeness of $(f_j)_{j \in J}$ in $L^2(\mathbb{P})$ implies that

$$\|h\|_{2,\mathbb{P}}^2 = \sum_{j \in J} \langle I_k h, f_j \rangle_{\mathbb{P}}^2 = \sum_{j \in J} \langle h, S_k h_j \rangle^2 = \sum_{j \in J} \lambda_j \langle h, h_j \rangle^2.$$

Moreover, continuity of h implies that $\|h\|_{2,\mathbb{P}}^2 = 0$ if and only if $h \equiv 0$ on $\mathcal{X}_o$. Hence,

$$\{k(x_o, \cdot) : x_o \in \mathcal{X}_o\}^\perp = \{h_j : j \in J\}^\perp,$$

and this is equivalent to $\overline{\text{span}}(h_j : j \in J) = \overline{\text{span}}(k(x_o, \cdot) : x_o \in \mathcal{X}_o)$. Consequently, any function $h \in \overline{\text{span}}(k(x_o, \cdot) : x_o \in \mathcal{X}_o)$ may be written as

$$h = \sum_{j \in J} \langle h, h_j \rangle h_j$$

with unconditional convergence in $\mathcal{H}$ and unconditional pointwise convergence in $\mathbb{R}$. In particular, for $x \in \mathcal{X}_o$ and $y \in \mathcal{X}$,

$$k(x, y) = \sum_{j \in J} \langle k(x, \cdot), h_j \rangle h_j(y) = \sum_{j \in J} h_j(x) h_j(y) = \sum_{j \in J} \lambda_j f_j(x) f_j(y).$$

□

Appendix: Auxiliary results from Functional Analysis and Probability Theory

A Compact operators on Hilbert spaces

Let $(\mathbb{X}, \|\cdot\|)$ and $(\mathbb{Y}, \|\cdot\|)$ be Banach spaces. (There is no need to distinguish the norms on $\mathbb{X}$ and $\mathbb{Y}$ notationally.) The closed unit ball $\{x \in \mathbb{X} : \|x\| \leq 1\}$ of $\mathbb{X}$ is denoted by $\mathbb{B}(\mathbb{X})$. Recall the definition of a compact linear operator.

Definition A.1. A linear operator $A : \mathbb{X} \rightarrow \mathbb{Y}$ is called compact, if for any bounded sequence $(x_n)_{n \geq 1}$, there exists a subsequence $(x_{n(k)})_{k \geq 1}$ such that $(Ax_{n(k)})_{k \geq 1}$ is a Cauchy sequence.

Equivalently, the image $\{Ax : x \in \mathbb{B}(\mathbb{X})\}$ is precompact, that is, for any fixed $\epsilon > 0$, it can be covered with finitely many balls of radius ϵ .

From now on, we consider a Hilbert space $\mathbb{X} = (\mathbb{X}, \langle \cdot, \cdot \rangle, \|\cdot\|)$. In this special case, compactness of an operator can be characterized differently. A useful notion in that context is weak convergence of sequences in $\mathbb{X}$.

Definition A.2 (Weak convergence). A sequence $(x_n)_{n \geq 1}$ in $\mathbb{X}$ converges weakly to a point $x \in \mathbb{X}$ if for any fixed $w \in \mathbb{X}$,

$$\lim_{n \rightarrow \infty} \langle x_n, w \rangle = \langle x, w \rangle.$$

Before going on, let us recall some basic facts about orthonormal systems in Hilbert spaces: A tuple $(\phi_j)_{j \in J}$ of vectors $\phi_j \in \mathbb{X}$ is called an *orthonormal system* if

$$\langle \phi_i, \phi_j \rangle = 1_{[i=j]} \quad \text{for all } i, j \in J.$$

Let us assume, for simplicity, that J is finite or countable. Then for any tuple $(\alpha_j)_{j \in J}$ of scalars with $\sum_{j \in J} |\alpha_j|^2 < \infty$,

$$x_o := \sum_{j \in J} \alpha_j \phi_j$$

defines a point $x_o \in \overline{\text{span}}(\phi_j : j \in J)$ with squared norm $\|x_o\|^2 = \sum_{j \in J} |\alpha_j|^2$. (In case of infinite J , the sum $\sum_{j \in J} \alpha_j \phi_j$ is meant as the unconditional limit of a series in $\mathbb{X}$.) For $y_o = \sum_{j \in J} \beta_j \phi_j$ with another tuple $(\beta_j)_{j \in J}$ of scalars such that $\sum_{j \in J} |\beta_j|^2 < \infty$, we have Parseval's identity,

$$\langle x_o, y_o \rangle = \sum_{j \in J} \alpha_j \bar{\beta}_j.$$

For $x \in \mathbb{X}$, the sum $\sum_{j \in J} |\langle x, \phi_j \rangle|^2$ is finite, and

$$\sum_{j \in J} \langle x, \phi_j \rangle \phi_j$$

is the orthogonal projection of x onto $\overline{\text{span}}(\phi_j : j \in J)$.

Finally, an orthonormal system $(\phi_j)_{j \in J}$ is called *complete*, if $\overline{\text{span}}(\phi_j : j \in J) = \mathbb{X}$. Some authors call a complete orthonormal system an orthonormal (Hilbert) basis of $\mathbb{X}$. The metric space $\mathbb{X}$ is separable if and only if it admits a finite or countable complete orthonormal system.

Lemma A.3 (Sequential weak compactness of $\mathbb{B}(\mathbb{X})$). *Let $(x_n)_{n \geq 1}$ be a sequence in the closed unit ball $\mathbb{B}(\mathbb{X})$ of the Hilbert space $\mathbb{X}$. Then there exist a subsequence $(x_{n(k)})_{k \geq 1}$ of $(x_n)_{n \geq 1}$ converging weakly to a point $x \in \mathbb{B}(\mathbb{X})$.*

Proof of Lemma A.3. Applying the Gram–Schmidt procedure to $(x_n)_{n \geq 1}$ yields an orthonormal system $(\phi_j)_{j \in J}$ in $\mathbb{X}$ such that $\text{span}(x_n : n \geq 1) \subset \text{span}(\phi_j : j \in J)$, where J is finite or countable. Now we can apply Cantor's diagonalization method to show that for some subsequence $(x_{n(k)})_{k \geq 1}$ of $(x_n)_{n \geq 1}$,

$$\gamma_j := \lim_{k \rightarrow \infty} \langle x_{n(k)}, \phi_j \rangle$$

exists for all $j \in J$. In particular, by Fatou's lemma and Parseval's identity,

$$\sum_{j \in J} |\gamma_j|^2 = \sum_{j \in J} \lim_{k \rightarrow \infty} |\langle x_{n(k)}, \phi_j \rangle|^2 \leq \liminf_{k \rightarrow \infty} \sum_{j \in J} |\langle x_{n(k)}, \phi_j \rangle|^2 = \liminf_{k \rightarrow \infty} \|x_{n(k)}\|^2 \leq 1.$$

Consequently, the point

$$x := \sum_{j \in J} \gamma_j \phi_j$$

is well-defined in $\overline{\text{span}}(\phi_j : j \in J)$ and satisfies $\|x\| \leq 1$.

The next step is to show that $(x_{n(k)})_{k \geq 1}$ converges weakly to x , that is, for any fixed $w \in \mathbb{X}$,

$$\lim_{k \rightarrow \infty} \langle x_{n(k)} - x, w \rangle = 0.$$

Indeed, since $x_{n(k)} - x \in \overline{\text{span}}(\phi_j : j \in J)$ with $\|x_{n(k)} - x\| \leq 2$,

$$\langle x_{n(k)} - x, w \rangle = \sum_{j \in J} \langle x_{n(k)} - x, \phi_j \rangle \overline{\langle w, \phi_j \rangle},$$

and for any finite set $J_o \subset J$, the modulus of the right hand side is no larger than

$$\begin{aligned} & \sum_{j \in J_o} |\langle x_{n(k)} - x, \phi_j \rangle| \|w\| + \sqrt{\sum_{j \in J \setminus J_o} |\langle x_{n(k)} - x, \phi_j \rangle|^2 \sum_{j \in J \setminus J_o} |\langle w, \phi_j \rangle|^2} \\ & \leq \sum_{j \in J_o} |\langle x_{n(k)} - x, \phi_j \rangle| \|w\| + 2 \sqrt{\sum_{j \in J \setminus J_o} |\langle w, \phi_j \rangle|^2}. \end{aligned}$$

The right hand side converges to $2 \sqrt{\sum_{j \in J \setminus J_o} |\langle w, \phi_j \rangle|^2}$ as $k \rightarrow \infty$, which can be arbitrarily small for suitable finite $J_o \subset J$. $\square$

Now we can state alternative definitions of compactness of a linear operator.

Theorem A.4. *In case of a Hilbert space $\mathbb{X}$, the following four properties of a linear operator $A : \mathbb{X} \rightarrow \mathbb{Y}$ are equivalent:*

- (i) *The operator A is compact.*
- (ii) *The image of the closed unit ball, $\{Ax : x \in \mathbb{B}(\mathbb{X})\}$, is a compact subset of $\mathbb{Y}$.*
- (iii) *If $(x_n)_{n \geq 1}$ is a bounded¹ sequence in $\mathbb{X}$ with weak limit 0, then $(Ax_n)_{n \geq 1}$ converges to 0.*

Proof of Theorem A.4. Property (i) of A is clearly a consequence of property (ii).

Now suppose that A has property (i), and let $(x_n)_{n \geq 1}$ be a bounded sequence in $\mathbb{X}$ converging weakly to 0. There exists a subsequence $(x_{n(k)})_{k \geq 1}$ of $(x_n)_{n \geq 1}$ such that

$$\lim_{k \rightarrow \infty} \|Ax_{n(k)}\| = \limsup_{n \rightarrow \infty} \|Ax_n\|,$$

and by compactness of A , we may even assume that $y := \lim_{k \rightarrow \infty} Ax_{n(k)}$ exists. Consequently, $\limsup_{n \rightarrow \infty} \|Ax_n\| = \|y\|$, so it remains to show that $y = 0$. Indeed, for any continuous linear functional L on $\mathbb{Y}$,

$$L(y) = \lim_{k \rightarrow \infty} L(Ax_{n(k)}) = \lim_{k \rightarrow \infty} \langle x_{n(k)}, w \rangle = 0,$$

where $w \in \mathbb{X}$ is the unique vector such that the continuous linear functional $x \mapsto L(Ax)$ is given by $L(Ax) = \langle x, w \rangle$. Consequently, $L(y) = 0$ for any continuous linear functional L on $\mathbb{Y}$, and by the Hahn–Banach theorem, this is wellknown to be equivalent to $y = 0$.

¹According to the Banach–Steinhaus theorem, a weakly convergent sequence in a Hilbert space is automatically bounded, but we do not need this fancy result.

Finally, suppose that A has property (iii). Let $(y_n)_{n \geq 1}$ be an arbitrary sequence in $\{Ax : x \in \mathbb{B}(\mathbb{X})\}$. Writing $y_n = Ax_n$ with $x_n \in \mathbb{B}(\mathbb{X})$, Lemma A.3 implies the existence of a subsequence $(x_{n(k)})_{k \geq 1}$ of $(x_n)_{n \geq 1}$ converging weakly to a point $x \in \mathbb{B}(\mathbb{X})$. Hence $(x_{n(k)} - x)_{k \geq 1}$ is a bounded sequence with weak limit 0. By property (iii) of A , the sequence $(y_{n(k)} - Ax)_{k \geq 1} = (A(x_{n(k)} - x))_{k \geq 1}$ converges to 0. Hence, A has property (ii). $\square$

The previous results have several useful consequences. We just mention two of them, the first one for real or complex Hilbert spaces, the second one for real Hilbert spaces only.

Theorem A.5 (Singular value decomposition). *Let A be a continuous linear operator from the Hilbert space $\mathbb{X}$ to the Hilbert space $\mathbb{Y}$. The following two properties of A are equivalent:*

- (a) A is compact.
- (b) There exist orthonormal systems $(\phi_j)_{j \in J}$ in $\mathbb{X}$, $(\psi_j)_{j \in J}$ in $\mathbb{Y}$ and a bounded tuple $(\sigma_j)_{j \in J}$ of nonnegative real numbers such that for any $x \in \mathbb{X}$,

$$Ax = \sum_{j \in J} \sigma_j \langle x, \phi_j \rangle \psi_j.$$

The index set J is finite, or $J = \mathbb{N}$. In the latter case, $\lim_{j \rightarrow \infty} \sigma_j = 0$.

Remark A.6. If A satisfies conditions (a–b) in Theorem A.5, then

$$\begin{aligned} A\phi_j &= \sigma_j \psi_j \quad \text{for all } j \in J, \\ Ax &= 0 \quad \text{whenever } x \perp \{\phi_j : j \in J\}. \end{aligned}$$

The numbers σ_j , $j \in J$, are called singular values of A , and the vectors ϕ_j, ψ_j are the corresponding singular vectors. The proof of the theorem reveals that

$$\max_{j \in J} \sigma_j = \max_{x \in \mathbb{B}(\mathbb{X})} \|Ax\|.$$

Remark A.7 (Adjoint operators). Recall that the (Hilbert) adjoint operator of a continuous linear operator $A : \mathbb{X} \rightarrow \mathbb{Y}$ is the unique continuous linear operator $A^* : \mathbb{Y} \rightarrow \mathbb{X}$ such that

$$\langle Ax, y \rangle = \langle A^*y, x \rangle$$

for all $x \in \mathbb{X}$ and $y \in \mathbb{Y}$. If A satisfies conditions (a–b) in Theorem A.5, then its adjoint operator A^* is given by

$$A^*y = \sum_{j \in J} \sigma_j \langle y, \psi_j \rangle \phi_j.$$

In particular, A^* is also compact with the same singular values as A .

Proof of Theorem A.5. Suppose that A is compact. We only consider the nontrivial case that $A \neq 0$. Since $\|\cdot\|$ is a continuous function on $\mathbb{Y}$, and since $\{Ax : x \in \mathbb{B}(\mathbb{X})\}$ is a compact subset of $\mathbb{Y}$, there exists a vector $\phi_1 \in \mathbb{X}$ with $\|\phi_1\| \leq 1$ and

$$\|A\phi_1\| = \sigma_1 := \sup_{x \in \mathbb{B}(\mathbb{X})} \|Ax\|.$$

It follows from $A \neq 0$ that $\sigma_1 > 0$ and $\|\phi_1\| = 1$. We may write $A\phi_1 = \sigma_1\psi_1$ with the unit vector $\psi_1 := \sigma_1^{-1}A\phi_1$ in $\mathbb{Y}$.

Suppose that for some $j_o \in \mathbb{N}$, we have chosen orthonormal vectors $\phi_1, \dots, \phi_{j_o} \in \mathbb{X}$ and orthonormal vectors $\psi_1, \dots, \psi_{j_o} \in \mathbb{Y}$ such that for $1 \leq j \leq j_o$,

$$A\phi_j = \sigma_j\psi_j$$

with

$$\sigma_j := \sup_{x \in \mathbb{B}(\mathbb{X}_j)} \|Ax\| > 0, \quad \text{and} \quad \mathbb{X}_j := \begin{cases} \mathbb{X} & \text{if } j = 1, \\ \{\phi_1, \dots, \phi_{j-1}\}^\perp & \text{if } j > 1. \end{cases}$$

Since $\mathcal{X}_1 \supset \dots \supset \mathcal{X}_{j_o}$, this implies that $\sigma_1 \geq \dots \geq \sigma_{j_o}$. If $Ax = 0$ for all $x \in \mathcal{X}_{j_o+1} := \{\phi_1, \dots, \phi_{j_o}\}^\perp$, we set $J = \{1, \dots, j_o\}$. Otherwise, note first that $Ax \perp \{\psi_1, \dots, \psi_{j_o+1}\}$ for all $x \in \mathcal{X}_{j_o+1}$. Indeed, for each $j \in \{1, \dots, j_o\}$ and any scalar t (real or complex), it follows from $\phi_j + tx \in \mathbb{X}_j$ that

$$\sigma_j^2 \geq \frac{\|A(\phi_j + tx)\|^2}{\|\phi_j + tx\|^2} = \frac{\|\sigma_j\psi_j + tAx\|^2}{1 + |t|^2\|x\|^2} = \frac{\sigma_j^2 + 2\sigma_j \operatorname{Re}(t\langle Ax, \psi_j \rangle) + |t|^2}{1 + |t|^2\|x\|^2},$$

and the right-hand side equals

$$\sigma_j^2 + 2\sigma_j \operatorname{Re}(t\langle Ax, \psi_j \rangle) + o(|t|^2)$$

as $t \rightarrow 0$. Consequently, $\langle Ax, \psi_j \rangle = 0$ for all $j \in \{1, \dots, j_o\}$. Hence, we may consider A temporarily as a compact linear operator from $\mathcal{X}_{j_o+1}$ to $\{\psi_1, \dots, \psi_{j_o}\}^\perp$, and there exist unit vectors $\phi_{j_o+1} \in \mathcal{X}_{j_o+1}$ and $\psi_{j_o+1} \in \{\psi_1, \dots, \psi_{j_o}\}^\perp$ such that

$$A\psi_{j_o+1} = \sigma_{j_o+1}\psi_{j_o+1}$$

with

$$\sigma_{j_o+1} := \sup_{x \in \mathbb{X}_{j_o+1}: \|x\| \leq 1} \|Ax\| > 0.$$

Inductively, we obtain orthonormal systems $(\phi_j)_{j \in J}$ in $\mathbb{X}$ and $(\psi_j)_{j \in J}$ in $\mathbb{Y}$, where either $J = \{1, \dots, j_o\}$ for some $j_o \in \mathbb{N}$, or $J = \mathbb{N}$. Moreover, $A\phi_j = \sigma_j\psi_j$ for $j \in J$, where $\sigma_j > 0$ is non-increasing in $j \in J$. In case of $J = \mathbb{N}$, the sequence $(\sigma_j)_{j \geq 1}$ has to converge to 0. Indeed, if $\sigma_j \geq \delta > 0$ for all $j \geq 0$, then $\|A\phi_j - A\phi_{j'}\|^2 = \sigma_j^2 + \sigma_{j'}^2 \geq 2\delta^2$ for different $j, j' \in \mathbb{N}$. Thus $(A\phi_j)_{j \geq 1}$ would not contain a Cauchy subsequence, a contradiction to compactness of A . For finite or infinite J , our construction implies that $Ax = 0$ for any $x \in \{\phi_j : j \in J\}^\perp$. In case of a finite set J , this is obvious. In case of $J = \mathbb{N}$, if $\|Ax\| > 0$ for some unit vector x in $\{\phi_j : j \in J\}^\perp$, then $\sigma_j \geq \|Ax\|$ for all $j \in \mathbb{N}$, again a contradiction to $\lim_{j \rightarrow \infty} \sigma_j = 0$. For an arbitrary point $x \in \mathbb{X}$ and its orthogonal projection $x_o = \sum_{j \in J} \langle x, \phi_j \rangle \phi_j$ onto $\overline{\operatorname{span}}(\phi_j : j \in J)$, $Ax = Ax_o$ is given by $Ax = Ax_o = \sum_{j \in J} \langle x, \phi_j \rangle A\phi_j = \sum_{j \in J} \sigma_j \langle x, \phi_j \rangle \psi_j$.

So far we have shown that property (a) of A implies property (b). Now suppose that A has property (b). Let $(x_n)_{n \geq 1}$ be a bounded sequence in $\mathbb{X}$ converging weakly to 0. With $D :=$

$\sup_{n \geq 1} \|x_n\| < \infty$, for any fixed finite set $J_o \subset J$,

$$\begin{aligned}
\|Ax_n\|^2 &= \sum_{j \in J} \sigma_j^2 |\langle x_n, \phi_j \rangle|^2 \\
&\leq \max_{i \in J_o} \sigma_i^2 \sum_{j \in J_o} |\langle x_n, \phi_j \rangle|^2 + \sup_{i \in J \setminus J_o} \sigma_i^2 \sum_{j \in J \setminus J_o} |\langle x_n, \phi_j \rangle|^2 \\
&\leq \max_{i \in J_o} \sigma_i^2 \sum_{j \in J_o} |\langle x_n, \phi_j \rangle|^2 + D^2 \sup_{i \in J \setminus J_o} \sigma_i^2 \\
&\rightarrow D^2 \sup_{i \in J \setminus J_o} \sigma_i^2 \quad \text{as } n \rightarrow \infty,
\end{aligned}$$

where $\sup_{i \in \emptyset} \sigma_i^2 := 0$. Since $\sup_{i \in J \setminus J_o} \sigma_i^2$ can be arbitrarily small for suitable finite $J_o \subset J$, this shows that $(Ax_n)_{n \geq 1}$ converges to 0. Hence, by Theorem A.4, A is compact. $\square$

Theorem A.8 (Spectral decomposition). *Let A be a continuous and self-adjoint² linear operator from the real Hilbert space $\mathbb{X}$ to itself. Then the following two properties of A are equivalent:*

(a) A is compact.

(b) There exist an orthonormal system $(\phi_j)_{j \in J}$ in $\mathbb{X}$ and a bounded tuple $(\lambda_j)_{j \in J}$ of real numbers such that for any $x \in \mathbb{X}$,

$$Ax = \sum_{j \in J} \lambda_j \langle x, \phi_j \rangle \phi_j.$$

The index set J is finite, or $J = \mathbb{N}$. In the latter case, $\lim_{j \rightarrow \infty} \lambda_j = 0$.

Remark A.9. If A satisfies conditions (a–b) in Theorem A.8, then

$$\begin{aligned}
A\phi_j &= \lambda_j \phi_j \quad \text{for all } j \in J, \\
Ax &= 0 \quad \text{whenever } x \perp \{\phi_j : j \in J\}.
\end{aligned}$$

Moreover, A is positive semidefinite³ if and only if $\lambda_j \geq 0$ for all $j \in J$. It is positive definite⁴ if and only if $(\phi_j)_{j \in J}$ is complete in $\mathbb{X}$ and $\lambda_j > 0$ for all $j \in J$. These statements follow immediately from the representation

$$\langle Ax, x \rangle = \sum_{j \in J} \lambda_j \langle x, \phi_j \rangle^2$$

for arbitrary $x \in \mathbb{X}$.

Proof of Theorem A.8. We proceed similarly as in the proof of Theorem A.5, assuming that $A \neq 0$. Let $(x_n)_{n \geq 1}$ be a sequence in $\mathbb{B}(\mathbb{X})$ such that

$$\lim_{n \rightarrow \infty} |\langle x_n, Ax_n \rangle| = \sigma_1 := \sup_{x \in \mathbb{B}(\mathbb{X})} |\langle x, Ax \rangle|.$$

² $\langle Ax, y \rangle = \langle x, Ay \rangle$ for all $x, y \in \mathbb{X}$

³ $\langle Ax, x \rangle \geq 0$ for all $x \in \mathbb{X}$

⁴ $\langle Ax, x \rangle > 0$ for all $x \in \mathbb{X} \setminus \{0\}$

Note that $\sigma_1 > 0$, because otherwise, $\langle x, Ax \rangle = 0$ for arbitrary $x \in \mathbb{X}$, and this would imply that

$$0 = \langle x + y, A(x + y) \rangle = \langle x, Ax \rangle + \langle y, Ay \rangle + 2\langle Ax, y \rangle = 2\langle Ax, y \rangle$$

for arbitrary $x, y \in \mathbb{X}$, so $A \equiv 0$. By Lemma A.3, we may assume without loss of generality that $(x_n)_{n \geq 1}$ converges weakly to a point $\phi_1 \in \mathbb{B}(\mathbb{X})$, and Theorem A.4 shows that $(Ax_n)_{n \geq 1}$ converges to $A\phi_1$. Consequently,

$$|\langle x_n, Ax_n \rangle - \langle \phi_1, A\phi_1 \rangle| \leq \|Ax_n - A\phi_1\| + |\langle x_n - \phi_1, Ax \rangle| \rightarrow 0$$

as $n \rightarrow \infty$. Hence, $|\langle \phi_1, A\phi_1 \rangle| = \sigma_1$, and this implies that $\|\phi_1\| = 1$. Now we show that $A\phi_1 = \lambda_1 \phi_1$ for some real number λ_1 , so that $\sigma_1 = |\lambda_1|$, and that $Ax \perp \phi_1$ whenever $x \perp \phi_1$. Indeed, for any $x \perp \phi_1$ and $t \in \mathbb{R}$,

$$\sigma_1 \geq \frac{|\langle \phi_1 + tx, A(\phi_1 + tx) \rangle|}{\|\phi_1 + tx\|^2} = \frac{|\langle \phi_1, A\phi_1 \rangle + 2t\langle x, A\phi_1 \rangle + t^2\langle x, Ax \rangle|}{1 + t^2\|x\|^2},$$

and as $t \rightarrow 0$, the right hand side equals

$$\sigma_1 + 2 \operatorname{sign}(\langle \phi_1, A\phi_1 \rangle) \langle x, A\phi_1 \rangle t + o(t^2).$$

Consequently, $\langle x, A\phi_1 \rangle = 0$. Hence, $A\phi_1 \perp \phi_1^\perp$, and this is equivalent to saying that $A\phi_1 = \lambda_1 \phi_1$ for some $\lambda_1 \in \mathbb{R}$. When writing $\langle x, A\phi_1 \rangle = \langle Ax, \phi_1 \rangle$, we see that $Ax \perp \phi_1$ for all $x \perp \phi_1$.

Now we may treat A as a compact and self-adjoint linear operator from $\mathbb{X}_2 := \phi_1^\perp$ into $\mathbb{X}_2$. Inductively, we obtain an orthonormal system $(\phi_j)_{j \in J}$ in $\mathbb{X}$ with $J = \{1, \dots, j_o\}$ for some $j_o \in \mathbb{N}$ or $J = \mathbb{N}$. Moreover, for each $j \in J$,

$$A\phi_j = \lambda_j \phi_j$$

with a real number λ_j such that

$$|\lambda_j| = \sigma_j := \sup_{x \in \mathbb{B}(\mathbb{X}_j)} |\langle x, Ax \rangle| > 0,$$

where $\mathbb{X}_1 := \mathbb{X}$ and $\mathbb{X}_j := \{\phi_1, \dots, \phi_{j-1}\}^\perp$ if $j > 1$. This implies that σ_j is nonincreasing in $j \in J$. In case of $J = \mathbb{N}$, $\lim_{j \rightarrow \infty} \sigma_j = 0$, because otherwise $(A\phi_j)_{j \geq 1}$ would not contain a Cauchy subsequence. As in the proof of Theorem A.5, one can deduce that $Ax = 0$ for any vector $x \perp \{\phi_j : j \in J\}$. This implies the representation $Ax = \sum_{j \in J} \lambda_j \langle x, \phi_j \rangle \phi_j$ for arbitrary $x \in \mathbb{X}$. $\square$

B Hilbert–Schmidt operators

B.1 Some general theory

Let $\mathbb{X} = (\mathbb{X}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ and $\mathbb{Y} = (\mathbb{Y}, \langle \cdot, \cdot \rangle, \|\cdot\|)$ be separable Hilbert spaces.

Theorem B.1. For a continuous linear operator $A : \mathbb{X} \rightarrow \mathbb{Y}$, the following two properties are equivalent:

(a) For a given complete orthonormal system $(\phi_i)_{i \in I}$ of $\mathbb{X}$,

$$\|A\|_{\text{HS}} := \sqrt{\sum_{i \in I} \|A\phi_i\|^2} < \infty.$$

(b) The operator A is compact, and the tuple $(\sigma_j)_{j \in J}$ of its singular values, as described in Theorem A.5, satisfies

$$\|A\|_{\text{SVD}} := \sqrt{\sum_{j \in J} \sigma_j^2} < \infty.$$

The number $\|A\|_{\text{HS}} \in [0, \infty]$ does not depend on the particular choice of the complete orthonormal system $(\phi_i)_{i \in I}$. If A has properties (a–b), then $\|A\|_{\text{HS}} = \|A\|_{\text{SVD}}$.

Definition B.2 (Hilbert–Schmidt operator). A linear operator A with the properties (a–b) in Theorem B.1 is called a *Hilbert–Schmidt operator*.⁵ The number $\|A\|_{\text{HS}}$ is called the *Hilbert–Schmidt norm* of A .

Proof of Theorem B.1. Let us first verify that for any continuous linear operator A , the number $\|A\|_{\text{HS}} \in [0, \infty]$ does not depend on the particular choice of the complete orthonormal system $(\phi_i)_{i \in I}$ of $\mathbb{X}$. To this end, let $A^* : \mathbb{Y} \rightarrow \mathbb{X}$ be the (Hilbert) adjoint operator of A , that means, $\langle Ax, y \rangle = \langle x, A^*y \rangle$ for all $x \in \mathbb{X}$ and $y \in \mathbb{Y}$. Further, let $(\psi_\ell)_{\ell \in L}$ be a complete orthonormal system in $\mathbb{Y}$. Then

$$\|A\|_{\text{HS}}^2 = \sum_{i \in I} \sum_{\ell \in L} |\langle A\phi_i, \psi_\ell \rangle|^2 = \sum_{\ell \in L} \sum_{i \in I} |\langle A^*\psi_\ell, \phi_i \rangle|^2 = \sum_{\ell \in L} \|A^*\psi_\ell\|^2,$$

and the right-hand side does not depend on the particular choice of $(\phi_i)_{i \in I}$. (Noting that $(A^*)^* = A$, this calculation also shows that $\|A^*\|_{\text{HS}} = \|A\|_{\text{HS}}$.)

Suppose that A has property (a). Let $(x_n)_{n \geq 1}$ be a bounded sequence in $\mathbb{X}$ converging weakly to 0. With $D := \sup_{n \geq 1} \|x_n\| < \infty$, for any finite set $I_o \subset I$,

$$\begin{aligned} \|Ax_n\| &\leq \sum_{i \in I} |\langle x_n, \phi_i \rangle| \|A\phi_i\| \\ &\leq \|A\|_{\text{HS}} \sum_{i \in I_o} |\langle x_n, \phi_i \rangle| + \sqrt{\sum_{i \in I \setminus I_o} |\langle x_n, \phi_i \rangle|^2} \sum_{i \in I \setminus I_o} \|A\phi_i\|^2 \\ &\leq \|A\|_{\text{HS}} \sum_{i \in I_o} |\langle x_n, \phi_i \rangle| + D \sqrt{\sum_{i \in I \setminus I_o} \|A\phi_i\|^2} \\ &\rightarrow D \sqrt{\sum_{i \in I \setminus I_o} \|A\phi_i\|^2} \quad \text{as } n \rightarrow \infty. \end{aligned}$$

⁵Some authors restrict this definition to the case $\mathbb{X} = \mathbb{Y}$.

The right hand side becomes arbitrarily small for suitable $I_o \subset I$. Hence $(Ax_n)_{n \geq 1}$ converges to 0, showing that A is compact.

As shown before, the number $\|A\|_{\text{HS}}$ does not depend on the choice of $(\phi_i)_{i \in I}$. Now consider an arbitrary compact linear operator $A : \mathbb{X} \rightarrow \mathbb{Y}$. Consider a singular value decomposition of A as in Theorem A.5. If $(\phi_j)_{j \in J}$ is not complete in $\mathbb{X}$, we may complement it with a complete orthonormal system of the closed subspace $\{\phi_j : j \in J\}^\perp$ of $\mathbb{X}$ to get a complete orthonormal system $(\phi_i)_{i \in I}$ in $\mathbb{X}$, where $J \subset I$. Then $A\phi_i = \sigma_i \psi_i$ for $i \in J$, and $A\phi_i = 0$ for $i \in I \setminus J$, so

$$\|A\|_{\text{HS}}^2 = \sum_{j \in J} \sigma_j^2 = \|A\|_{\text{SVD}}^2.$$

This shows that properties (a) and (b) are equivalent, indeed. $\square$

B.2 The special case of square-integrable functions

Now let $(\mathcal{X}, \mathcal{A}, \mathbb{P})$ and $(\mathcal{Y}, \mathcal{B}, \mathbb{Q})$ be σ -finite measure spaces such that the corresponding real Hilbert spaces $\mathbb{X} = L^2(\mathbb{P})$ and $\mathbb{Y} = L^2(\mathbb{Q})$ are separable. Hilbert–Schmidt operators from $L^2(\mathbb{P})$ to $L^2(\mathbb{Q})$ can always be represented as integral operators as explained in the next two theorems.

Theorem B.3 (Integral operators). *Let $k \in \mathcal{L}^2(\mathbb{P} \otimes \mathbb{Q})$. For $f \in L^2(\mathbb{P})$ and $y \in \mathcal{Y}$ let*

$$T_k f(y) := \int_{\mathcal{X}} k(x, y) f(x) \mathbb{P}(dx)$$

if $\|k(\cdot, y)\|_{2, \mathbb{P}} < \infty$, and $T_k f(y) := 0$ else. This defines a Hilbert–Schmidt operator $T_k : L^2(\mathbb{P}) \rightarrow L^2(\mathbb{Q})$ with $\|T_k\|_{\text{HS}} = \|k\|_{2, \mathbb{P} \otimes \mathbb{Q}}$.

Remark B.4 (Essential uniqueness of the kernel k). *If $k, h \in \mathcal{L}^2(\mathbb{P} \otimes \mathbb{Q})$ such that $T_h \equiv T_k$, then $h - k \in \mathcal{L}^2(\mathbb{P} \otimes \mathbb{Q})$ satisfies $\|h - k\|_{2, \mathbb{P} \otimes \mathbb{Q}} = \|T_{h-k}\|_{\text{HS}} = \|T_h - T_k\|_{\text{HS}} = 0$, whence $h = k$ almost everywhere w.r.t. $\mathbb{P} \otimes \mathbb{Q}$.*

In what follows, for functions $f : \mathcal{X} \rightarrow \mathbb{R}$ and $g : \mathcal{Y} \rightarrow \mathbb{R}$, the function $(x, y) \mapsto f(x)g(y)$ on $\mathcal{X} \times \mathcal{Y}$ is denoted by $f \otimes g$. By Fubini’s theorem, for $f_1, f_2 \in L^2(\mathbb{P})$ and $g_1, g_2 \in L^2(\mathbb{Q})$,

$$\langle f_1 \otimes g_1, f_2 \otimes g_2 \rangle_{\mathbb{P} \otimes \mathbb{Q}} = \langle f_1, f_2 \rangle_{\mathbb{P}} \langle g_1, g_2 \rangle_{\mathbb{Q}}.$$

In particular, if $(f_i)_{i \in I}$ and $(g_j)_{j \in J}$ are orthonormal systems in $L^2(\mathbb{P})$ and $L^2(\mathbb{Q})$, respectively, then $(f_i \otimes g_j)_{(i,j) \in I \times J}$ is an orthonormal system in $L^2(\mathbb{P} \otimes \mathbb{Q})$.

Theorem B.5. *Let $A : L^2(\mathbb{P}) \rightarrow L^2(\mathbb{Q})$ be a Hilbert–Schmidt operator. Then there exists a unique function $k \in L^2(\mathbb{P} \otimes \mathbb{Q})$ such that $A = T_k$ with the integral operator T_k defined in Theorem B.3. Precisely, there exist a finite or countable set J , orthonormal systems $(f_j)_{j \in J}$ in $L^2(\mathbb{P})$, $(g_j)_{j \in J}$ in $L^2(\mathbb{Q})$ and a tuple $(\sigma_j)_{j \in J}$ of nonnegative real numbers such that $\sum_{j \in J} \sigma_j^2 = \|k\|_{2, \mathbb{P} \otimes \mathbb{Q}}^2$ and*

$$k = \sum_{j \in J} \sigma_j f_j \otimes g_j$$

Remark B.6. For A and k as in Theorem B.5,

$$\begin{aligned} Af_j &= \sigma_j g_j \quad \text{for all } j \in J, \\ Af &= 0 \quad \text{for all } f \perp \{f_j : j \in J\}. \end{aligned}$$

Proof of Theorem B.3. Note first that by Fubini's theorem,

$$\|k\|_{2,\mathbb{P} \otimes \mathbb{Q}}^2 = \int_{\mathcal{Y}} \|k(\cdot, y)\|_{2,\mathbb{P}}^2 \mathbb{Q}(dy).$$

Consequently, the set $N := \{y \in \mathcal{Y} : \|k(\cdot, y)\|_{2,\mathbb{P}} = \infty\}$ satisfies $\mathbb{Q}(N) = 0$. For $f \in L^2(\mathbb{P})$ and $y \in \mathcal{Y} \setminus N$, it follows from the Cauchy–Schwarz inequality that

$$\int |k(x, y)f(x)| \mathbb{P}(dx) \leq \|k(\cdot, y)\|_{2,\mathbb{P}} \|f\|_{2,\mathbb{P}},$$

so $T_k f(y)$ is well-defined in $\mathbb{R}$, and

$$\|T_k f\|_{2,\mathbb{Q}}^2 = \int T_k f(y)^2 \mathbb{Q}(dy) \leq \|k\|_{2,\mathbb{P} \otimes \mathbb{Q}}^2 \|f\|_{2,\mathbb{P}}^2.$$

If $(f_i)_{i \in I}$ is a complete orthonormal system in $L^2(\mathbb{P})$, then

$$\begin{aligned} \|T_k\|_{\text{HS}}^2 &= \sum_{i \in I} \|T_k f_i\|_{2,\mathbb{Q}}^2 \\ &= \sum_{i \in I} \int_{\mathcal{Y} \setminus N} \langle k(\cdot, y), f_i \rangle_{\mathbb{P}}^2 \mathbb{Q}(dy) \\ &= \int_{\mathcal{Y} \setminus N} \sum_{i \in I} \langle k(\cdot, y), f_i \rangle_{\mathbb{P}}^2 \mathbb{Q}(dy) \\ &= \int_{\mathcal{Y} \setminus N} \|k(\cdot, y)\|_{2,\mathbb{P}}^2 \mathbb{Q}(dy) \\ &= \|k\|_{2,\mathbb{P} \otimes \mathbb{Q}}^2. \end{aligned}$$

□

Proof of Theorem B.5. Uniqueness of the kernel function k is a consequence of Remark B.4. That any Hilbert–Schmidt operator A can be represented as integral operator T_k with a particular function k as stated follows directly from Theorems A.5 and B.1. Indeed, there exist a finite or countable set J , orthonormal systems $(f_j)_{j \in J}$ in $L^2(\mathbb{P})$, $(g_j)_{j \in J}$ in $L^2(\mathbb{Q})$ and a tuple $(\sigma_j)_{j \in J}$ of nonnegative real numbers such that $\sum_{j \in J} \sigma_j^2 < \infty$ and

$$Af = \sum_{j \in J} \sigma_j \langle f, f_j \rangle_{\mathbb{P}} g_j$$

for any $f \in L^2(\mathbb{P})$. Since $\sum_{j \in J} \sigma_j^2 < \infty$, and since $(f_j \otimes g_j)_{j \in J}$ is an orthonormal system in $L^2(\mathbb{P} \otimes \mathbb{Q})$, the function $k := \sum_{j \in J} \sigma_j f_j \otimes g_j$ is well-defined in $L^2(\mathbb{P} \otimes \mathbb{Q})$ and satisfies

$\|k\|_{2, \mathbb{P} \otimes \mathbb{Q}}^2 = \sum_{j \in J} \sigma_j^2 = \|A\|_{\text{HS}}^2$. Moreover, for any $g \in L^2(\mathbb{Q})$,

$$\begin{aligned} \langle Af, g \rangle_{\mathbb{Q}} &= \sum_{j \in J} \sigma_j \langle f, f_j \rangle_{\mathbb{P}} \langle g_j, g \rangle_{\mathbb{Q}} \\ &= \sum_{j \in J} \sigma_j \langle f \otimes g, f_j \otimes g_j \rangle_{\mathbb{P} \otimes \mathbb{Q}} \\ &= \langle f \otimes g, k \rangle_{\mathbb{P} \otimes \mathbb{Q}} \\ &= \int_{\mathcal{X}} \int_{\mathcal{Y}} f(x) g(y) k(x, y) \mathbb{P}(dx) \mathbb{Q}(dy) \\ &= \langle T_k f, g \rangle_{\mathbb{Q}}. \end{aligned}$$

This shows that $A = T_k$. □

We end this section with a result about selfadjoint Hilbert–Schmidt operators $A : L^2(\mathbb{P}) \rightarrow L^2(\mathbb{P})$.

Theorem B.7. *Let $A : L^2(\mathbb{P}) \rightarrow L^2(\mathbb{P})$ be a Hilbert–Schmidt operator. A is self-adjoint if and only if $A = T_k$ for some symmetric kernel $k \in \mathcal{L}^2(\mathbb{P} \otimes \mathbb{P})$. In this case, there exist an orthonormal system $(f_j)_{j \in J}$ in $L^2(\mathbb{P})$ and a tuple $(\lambda_j)_{j \in J}$ of real numbers such that $\sum_{j \in J} \lambda_j^2 = \|k\|_{2, \mathbb{P} \otimes \mathbb{P}}^2$, and*

$$k = \sum_{j \in J} \lambda_j f_j \otimes f_j.$$

Moreover, A is positive semidefinite if and only if $\lambda_j \geq 0$ for all $j \in J$. And A is strictly positive definite if and only if $(f_j)_{j \in J}$ is complete in $L^2(\mathbb{P})$ and $\lambda_j > 0$ for all $j \in J$.

The proof of this result is essentially a consequence of Theorems A.8 and B.1, see also the proof of Theorem B.5.

B.3 Covariance kernels and operators

An important special case of Hilbert–Schmidt operators are covariance operators of random vectors X and Y , defined on a common probability space $(\Omega, \mathcal{A}, \mathbb{P})$, with values in separable real Hilbert spaces $\mathbb{X}$ and $\mathbb{Y}$, respectively. If $\mathbb{E} \|X\| < \infty$ and $\mathbb{E} \|Y\| < \infty$, then the mappings $v \mapsto \mathbb{E} \langle X, v \rangle$ and $w \mapsto \mathbb{E} \langle Y, w \rangle$ are continuous linear forms on $\mathbb{X}$ and $\mathbb{Y}$, respectively. Thus, by Riesz’ representation theorem, there exist unique vectors $\mu_X = \mathbb{E}(X) \in \mathbb{X}$ and $\mu_Y = \mathbb{E}(Y) \in \mathbb{Y}$ such that $\mathbb{E} \langle X, v \rangle = \langle \mu_X, v \rangle$ for all $v \in \mathbb{X}$, and $\mathbb{E} \langle Y, w \rangle = \langle \mu_Y, w \rangle$ for all $w \in \mathbb{Y}$.

From now on we assume that even $\mathbb{E}(\|X\|^2), \mathbb{E}(\|Y\|^2) < \infty$.

Definition B.8 (Covariance function). The covariance function $\text{Cov}_{XY} : \mathbb{X} \times \mathbb{Y}$ of X and Y is defined via

$$\begin{aligned} \text{Cov}_{XY}(v, w) &:= \text{Cov}(\langle X, v \rangle, \langle Y, w \rangle) \\ &= \mathbb{E}(\langle X - \mu_X, v \rangle \langle Y - \mu_Y, w \rangle) \\ &= \mathbb{E}(\langle X, v \rangle \langle Y, w \rangle) - \langle \mu_X, v \rangle \langle \mu_Y, w \rangle. \end{aligned}$$

The covariance function Cov_{XY} is a bilinear form on $\mathbb{X} \times \mathbb{Y}$, and by means of the Cauchy–Schwarz inequality for $\mathbb{X}$, $\mathbb{Y}$ and $L^2(\mathbb{P})$, one can show that

$$|\text{Cov}_{XY}(v, w)| \leq \sqrt{\mathbb{E}(\|X - \mu_X\|^2) \mathbb{E}(\|Y - \mu_Y\|^2)} \|v\| \|w\| \quad (\text{B.1})$$

for arbitrary $v \in \mathbb{X}$, $w \in \mathbb{Y}$.

Theorem B.9 (Covariance operator). *There exists a unique Hilbert–Schmidt operator $\Sigma_{XY} : \mathbb{X} \rightarrow \mathbb{Y}$ such that*

$$\text{Cov}_{XY}(x, y) = \langle \Sigma_{XY} x, y \rangle \quad \text{for all } x \in \mathbb{X}, y \in \mathbb{Y},$$

and

$$\|\Sigma_{XY}\|_{\text{HS}}^2 \leq \mathbb{E}(\|X - \mu_X\|^2) \mathbb{E}(\|Y - \mu_Y\|^2).$$

This operator Σ_{XY} is called the covariance operator of X and Y .

Proof of Theorem B.9. The function Cov_{XY} is a bilinear form on $\mathbb{X} \times \mathbb{Y}$. Inequality (B.1) implies that for any fixed $v \in \mathbb{X}$, $\text{Cov}_{XY}(v, w)$ is a continuous linear function of $w \in \mathbb{Y}$. By Riesz’ representation theorem, there exists a unique vector $\Sigma(v) \in \mathbb{X}$ such that $\text{Cov}_{XY}(v, w) = \langle \Sigma(v), w \rangle$ for all $w \in \mathbb{Y}$. Bilinearity of $\text{Cov}_{XY}(\cdot, \cdot)$ implies that $v \mapsto \Sigma(v)$ is a linear function on $\mathbb{X}$. It is continuous, because for any $v \in \mathbb{X}$,

$$\|\Sigma(v)\| = \max_{w \in \mathbb{B}(\mathbb{Y})} |\langle \Sigma(v), w \rangle| = \max_{w \in \mathbb{B}(\mathbb{Y})} |\text{Cov}_{XY}(v, w)| \leq C \|v\|$$

with $C := \sqrt{\mathbb{E}(\|X - \mu_X\|^2) \mathbb{E}(\|Y - \mu_Y\|^2)}$, according to (B.1).

Finally, let $(\phi_i)_{i \in I}$ and $(\psi_j)_{j \in J}$ be complete orthonormal systems in $\mathbb{X}$ and $\mathbb{Y}$, respectively. Then, with the centered random vectors $X_o := X - \mu_X$ and $Y_o := Y - \mu_Y$,

$$\begin{aligned} \|\Sigma\|_{\text{HS}}^2 &= \sum_{i \in I} \|\Sigma \phi_i\|^2 \\ &= \sum_{i \in I} \sum_{j \in J} \langle \Sigma \phi_i, \psi_j \rangle^2 \\ &= \sum_{i \in I} \sum_{j \in J} \mathbb{E}(\langle X_o, \phi_i \rangle \langle Y_o, \psi_j \rangle)^2 \\ &\leq \sum_{i \in I} \sum_{j \in J} \mathbb{E}(\langle X_o, \phi_i \rangle^2) \mathbb{E}(\langle Y_o, \psi_j \rangle^2) \\ &= \mathbb{E}\left(\sum_{i \in I} \langle X_o, \phi_i \rangle^2\right) \mathbb{E}\left(\sum_{j \in J} \langle Y_o, \psi_j \rangle^2\right) \\ &= \mathbb{E}(\|X_o\|^2) \mathbb{E}(\|Y_o\|^2). \end{aligned}$$

□

Principal components. In the special case that $\mathbb{X} = \mathbb{Y}$ and $X \equiv Y$, the covariance operator Σ_{XY} becomes the covariance operator $\Sigma_X := \Sigma_{XX}$ of X . It is a self-adjoint, positive semidefinite Hilbert–Schmidt operator. Consequently, there exists an orthonormal system $(\phi_j)_{j \in J}$, where $J =$

$\{1, \dots, \dim(\mathbb{X})\}$ if $\dim(\mathbb{X}) < \infty$, or $J = \mathbb{N}$, and a tuple $(\lambda_j)_{j \in J} \in [0, \infty)^J$ such that λ_j is non-increasing in $j \in J$,

$$\text{Cov}_{XX}(\phi_i, \phi_j) = 1_{[i=j]} \lambda_j \quad \text{for } i, j \in J,$$

and $\|\Sigma_X\|_{\text{HS}}^2 \leq \mathbb{E}(\|X - \mu_X\|^2)^2$. More precisely,

$$\sum_{j \in J} \lambda_j = \mathbb{E}(\|X - \mu_X\|^2).$$

The random variable

$$\langle X, \phi_j \rangle$$

is called the j -th (theoretical) principal component of X .

C Consistency of empirical means

For an integer $n \in \mathbb{N}$, let $X = X_1, X_2, \dots, X_n$ be stochastically independent, identically distributed random variables with distribution $\mathbb{P}$ on a separable real Hilbert space $(\mathbb{X}, \langle \cdot, \cdot \rangle, \|\cdot\|)$. We assume that $\mathbb{E}(\|X\|^2) < \infty$. A natural estimator of the mean $\mu := \mathbb{E}(X) \in \mathbb{X}$ is given by the sample mean

$$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^n X_i,$$

that is, the mean of the empirical distribution $\hat{\mathbb{P}}_n := n^{-1} \sum_{i=1}^n \delta_{X_i}$. Elementary calculations show that

$$\mathbb{E}(\hat{\mu}_n) = \mu \quad \text{and} \quad \Sigma_{\hat{\mu}_n} = n^{-1} \Sigma_X.$$

In particular,

$$\mathbb{E}(\|\hat{\mu}_n - \mu\|^2) = n^{-1} \mathbb{E}(\|X - \mu\|^2) = n^{-1} (\mathbb{E}(\|X\|^2) - \|\mu\|^2).$$

C.1 A sub-Gaussian exponential inequality

In the special case of $\|X\|$ being bounded almost surely, there is a concentration of measure phenomenon:

Theorem C.1. *Suppose that $\|X\| \leq c < \infty$ almost surely. For arbitrary $\eta \geq 0$,*

$$\left. \begin{aligned} &\mathbb{P}(\|\hat{\mu}_n - \mu\| \geq \mathbb{E}\|\hat{\mu}_n - \mu\| + \eta) \\ &\mathbb{P}(\|\hat{\mu}_n - \mu\| \leq \mathbb{E}\|\hat{\mu}_n - \mu\| - \eta) \end{aligned} \right\} \leq \exp(-n\eta^2/(2c^2)).$$

In particular, for any $\delta \in (0, 1]$,

$$\mathbb{P}(\|\hat{\mu}_n - \mu\| \geq c(1 + \sqrt{2 \log(1/\delta)})n^{-1/2}) \leq \delta.$$

Remark C.2 (Separable RKHSs). Recall that if $(\mathcal{X}, d)$ is a separable metric space, and if $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is a bounded and continuous positive definite kernel, then the corresponding RKHS $\mathcal{H} = \mathcal{H}_k$ is separable too. Now suppose that for some real constant $C > 0$, the kernel k satisfies $k \leq C$, which is equivalent to $\|k\|_\infty \leq C$. If $\tilde{X}_1, \tilde{X}_2, \tilde{X}_3, \dots$ are independent, identically distributed random variables with distribution $\tilde{\mathbb{P}}$ on (the Borel subsets of) $\mathcal{X}$, then $X_i := k(\tilde{X}_i, \cdot) \in \mathcal{H}$ defines independent, identically distributed random variables $X_i \in \mathcal{H}$, and

$$\|X_i\|_{\mathcal{H}} = k(\tilde{X}_i, \tilde{X}_i)^{1/2} \leq c := C^{1/2}.$$

Consequently, Theorem C.1 applies also to empirical kernel mean embeddings under the stated conditions on $\mathcal{X}$ and the kernel k (cf. Theorem 3.4 in Muandet et al., 2020).

Proof of Theorem C.1. The proof is essentially an adaptation of well-known inequalities due to Hoeffding, Azuma and McDiarmid. Recall first the classical Hoeffding inequality: For a random variable Y with values in a compact interval $[a, b]$,

$$\mathbb{E} \exp(t(Y - \mathbb{E}(Y))) \leq \exp(t^2(b - a)^2/8) \quad \text{for all } t \in \mathbb{R}. \quad (\text{C.1})$$

In our setting, for $1 \leq k \leq n$, let $S_k := \|n^{-1} \sum_{i=1}^k X_i - \mu\|$ and $\sigma_k := \mathbb{E}(S_k)$. Further, let $D_k := S_k - S_{k-1}$ and $\delta_k := \mathbb{E}(D_k)$, where $S_0 := \|\mu\| =: \sigma_0$. Then the triangular inequality for $\|\cdot\|$ yields that

$$|D_k| \leq \|X_k\|/n \leq c/n,$$

that is, $D_k \in [-c/n, c/n]$. Consequently, for $t \in \mathbb{R}$ and $1 \leq k \leq n$, Hoeffding's inequality (C.1) leads to

$$\begin{aligned} \mathbb{E} \exp(t(S_k - \sigma_k)) &= \mathbb{E}(\exp(t(D_k - \delta_k)) \exp(t(S_{k-1} - \sigma_{k-1}))) \\ &= \mathbb{E} \mathbb{E}(\exp(t(D_k - \delta_k)) \exp(t(S_{k-1} - \sigma_{k-1})) \mid (X_j)_{j < k}) \\ &= \mathbb{E}(\mathbb{E}(\exp(t(D_k - \delta_k)) \mid (X_j)_{j < k}) \exp(t(S_{k-1} - \sigma_{k-1}))) \\ &\leq \mathbb{E}(\exp(t^2 c^2 / (2n^2)) \exp(t(S_{k-1} - \sigma_{k-1}))) \\ &= \exp(t^2 c^2 / (2n^2)) \mathbb{E} \exp(t(S_{k-1} - \sigma_{k-1})). \end{aligned}$$

Since $S_0 - \sigma_0 \equiv 0$, applying this inequality for $k = n, n-1, \dots, 1$ leads to

$$\mathbb{E} \exp(t(S_n - \sigma_n)) \leq \exp(t^2 c^2 / (2n^2))^n = \exp(t^2 c^2 / (2n)).$$

In particular, for any $\eta \geq 0$,

$$\begin{aligned} \mathbb{P}(\pm(S_n - \sigma) \geq \eta) &\leq \inf_{t \geq 0} \mathbb{E} \exp(\pm t(S_n - \sigma_n) - t\eta) \\ &\leq \inf_{t \geq 0} \exp(t^2 c^2 / (2n) - t\eta) \\ &= \exp(-n\eta^2 / (2c^2)), \end{aligned}$$

where the optimal $t \geq 0$ equals $t = n\eta/c^2$. Since $S_n = \|\hat{\mu}_n - \mu\|$, this proves the first set of inequalities. The second set of inequalities is a simple consequence of the fact that

$$\sigma_n^2 = \mathbb{E}(S_n)^2 \leq \mathbb{E}(S_n^2) = n^{-1} \mathbb{E}(\|X - \mu\|^2) \leq n^{-1} \mathbb{E}(\|X\|^2) \leq n^{-1} c^2.$$

□

Proof of Hoeffding's inequality (C.1). Note first that

$$t(Y - \mathbb{E}(Y)) = t(b - a) \left(\frac{Y - a}{b - a} - \mathbb{E} \left(\frac{Y - a}{b - a} \right) \right),$$

so it suffices to consider the case that $Y \in [0, 1]$ and show that for any $t \in \mathbb{R}$,

$$\log \mathbb{E} \exp(t(Y - \mu)) \leq t^2/8,$$

where $\mu := \mathbb{E}(Y) \in [0, 1]$. By convexity of $y \mapsto \exp(ty)$,

$$\exp(tY) \leq (1 - Y) \exp(0) + Y \exp(t) = 1 - Y + Y e^t,$$

whence

$$\log \mathbb{E} \exp(t(Y - \mu)) = \log \mathbb{E} \exp(tY) - t\mu \leq h(t) := \log(1 - \mu + \mu e^t) - t\mu.$$

But $h(0) = 0$, and

$$\begin{aligned} h'(t) &= \frac{\mu e^t}{1 - \mu + \mu e^t} - \mu = 1 - \mu - \frac{1 - \mu}{1 - \mu + \mu e^t}, \\ h'(0) &= 0, \\ h''(t) &= \frac{(1 - \mu)\mu e^t}{(1 - \mu + \mu e^t)^2} = \frac{1 - \mu}{1 - \mu + \mu e^t} \left(1 - \frac{1 - \mu}{1 - \mu + \mu e^t} \right) \leq 1/4. \end{aligned}$$

Hence,

$$h(t) = h(0) + h'(0)t + h''(\xi(t))t^2/2 \leq t^2/8$$

for some $\xi(t)$ between 0 and t . □

C.2 A Central Limit Theorem

Theorem C.3 (Central Limit Theorem). *There exists an $\mathbb{X}$ -valued random variable G such that $\mathbb{E}(\|G\|^2) < \infty$ and*

$$\langle G, w \rangle \sim \mathcal{N}(0, \text{Var}(\langle X, w \rangle)) \quad \text{for all } w \in \mathbb{X}.$$

In particular, $\text{Cov}_{GG} \equiv \text{Cov}_{XX}$, $\mathbb{E}(\|G\|^2) = \mathbb{E}(\|X - \mu\|^2)$ and $\mathbb{E}(G) = 0$. Moreover, for any fixed continuous function $f : \mathbb{X} \rightarrow \mathbb{R}$ such that $\mathcal{X} \ni x \mapsto f(x)/(1 + \|x\|^2)$ is bounded,

$$\lim_{n \rightarrow \infty} \mathbb{E} f(\sqrt{n}(\hat{\mu}_n - \mu)) = \mathbb{E} f(G).$$

Remark C.4. The distribution of an $\mathbb{X}$ -valued random variable G is uniquely determined by the distribution of all random variables $\langle G, w \rangle$, w a unit vector in $\mathbb{X}$. This can be proved with similar arguments as in the proof of Proposition 5.7, working directly with characteristic functions instead of moment-generating functions.

Remark C.5. The proof of Theorem C.3 shows the following fact: Let $\Sigma : \mathbb{X} \rightarrow \mathbb{X}$ be a self-adjoint, positive semidefinite Hilbert–Schmidt operator with spectral decomposition

$$\Sigma = \sum_{j \in J} \lambda_j \langle \cdot, \phi_j \rangle \phi_j,$$

where $(\phi_j)_{j \in J}$ is a complete orthonormal system in $\mathbb{X}$, and the eigenvalues $\lambda_j \geq 0$ satisfy $\sum_{j \in J} \lambda_j < \infty$. (This is stronger than $\|\Sigma\|_{\text{HS}} < \infty$!) Then there exists a unique probability distribution on $\mathbb{X}$, denoted by $\mathcal{N}_{\mathbb{X}}(0, \Sigma)$, such that a random variable G with that distribution satisfies

$$\langle G, w \rangle \sim \mathcal{N}(0, \langle \Sigma w, w \rangle) \quad \text{for all } w \in \mathbb{X}.$$

In particular, $\mathbb{E}(\|G\|^2) = \sum_{j \in J} \lambda_j$.

The proof of Theorem C.3 uses an elementary lemma about weak convergence. Its proof is left to the reader as an exercise.

Lemma C.6. Let $(Q_n)_{n \geq 1}$ be a sequence of probability distributions on a metric space $\mathcal{X}$ which converges weakly to a distribution Q on $\mathcal{X}$. Let $h : \mathcal{X} \rightarrow [1, \infty)$ be a continuous and unbounded function. Suppose that $\int h dQ$ and $\int h dQ_n$ are finite for all $n \in \mathbb{N}$, and

$$\lim_{n \rightarrow \infty} \int h dQ_n = \int h dQ.$$

Then,

$$\lim_{n \rightarrow \infty} \int f dQ_n = \int f dQ$$

for any continuous function $f : \mathcal{X} \rightarrow \mathbb{R}$ such that f/h is bounded.

Proof of Theorem C.3. Let $\Sigma : \mathbb{X} \rightarrow \mathbb{X}$ be an arbitrary self-adjoint and positive semidefinite Hilbert–Schmidt operator with spectral decomposition $\Sigma = \sum_{j \in J} \lambda_j \langle \cdot, \phi_j \rangle \phi_j$, where $(\phi_j)_{j \in J}$ is a complete orthonormal system in $\mathbb{X}$ and $(\lambda_j)_{j \in J} \in [0, \infty)^J$ satisfies $\sum_{j \in J} \lambda_j < \infty$. If $(Z_j)_{j \in J}$ is a tuple of stochastically independent random variables with standard Gaussian distribution on $\mathbb{R}$, then

$$G := \sum_{j \in J} \sqrt{\lambda_j} Z_j \phi_j$$

is (almost surely) well-defined in $\mathbb{X}$, because

$$\mathbb{E}\left(\sum_{j \in J} \lambda_j Z_j^2\right) = \sum_{j \in J} \lambda_j < \infty.$$

Then, for any $w \in \mathbb{X}$,

$$\langle G, w \rangle = \sum_{j \in J} \sqrt{\lambda_j} Z_j \langle w, \phi_j \rangle,$$

and the sum or series on the right hand has a centered Gaussian distribution with variance

$$\sum_{j \in J} \lambda_j \langle w, \phi_j \rangle^2 = \langle \Sigma w, w \rangle.$$

In the special case of Σ being the covariance operator of $X \sim \mathbb{P}$, the random variable G has the desired Gaussian distribution on $\mathbb{X}$, and $\text{Cov}_{GG}(w, w) = \text{Cov}_{XX}(w, w)$ for all $w \in \mathbb{X}$. But this implies that $\text{Cov}_{GG} \equiv \text{Cov}_{XX}$, because any symmetric, bilinear function $\text{Cov} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ satisfies $4 \text{Cov}(v, w) = \text{Cov}(v + w, v + w) - \text{Cov}(v - w, v - w)$ for arbitrary $v, w \in \mathbb{X}$.

Note that for all $n \in \mathbb{N}$, the random variables X , G and $G_n := n^{1/2}(\hat{\mu}_n - \mu)$ have the same covariance function. This implies that for an arbitrary finite set $J_o \subset J$ and the orthogonal projection Π_o of $\mathbb{X}$ onto $\mathbb{X}_o := \text{span}(\phi_j : j \in J_o)$,

$$\mathbb{E}(\|G_n - \Pi_o G_n\|^2) = \mathbb{E}(\|G - \Pi_o G\|^2) = \epsilon(J_o) := \sum_{j \in J \setminus J_o} \lambda_j.$$

Moreover, it follows from the Central Limit Theorem in Euclidean spaces that $\Pi_o G_n$ converges in distribution to $\Pi_o G$ as $n \rightarrow \infty$. Consequently, for any fixed bounded, Lipschitz continuous function $f : \mathbb{X} \rightarrow \mathbb{R}$,

$$\begin{aligned} & |\mathbb{E}f(G_n) - \mathbb{E}f(G)| \\ & \leq \|f\|_{\text{Lip}} \mathbb{E}\|G_n - \Pi_o G_n\| + |\mathbb{E}f(\Pi_o G_n) - \mathbb{E}f(\Pi_o G)| + \|f\|_{\text{Lip}} \mathbb{E}\|G - \Pi_o G\| \\ & \leq 2\|f\|_{\text{Lip}} \epsilon(J_o)^{1/2} + |\mathbb{E}f(\Pi_o G_n) - \mathbb{E}f(\Pi_o G)| \\ & \rightarrow 2\|f\|_{\text{Lip}} \epsilon(J_o)^{1/2} \quad \text{as } n \rightarrow \infty, \end{aligned}$$

where $\|f\|_{\text{Lip}}$ is the supremum of $|f(x) - f(y)|/\|x - y\|$ over all different $x, y \in \mathbb{X}$. But $\epsilon(J_o)$ is arbitrarily small for suitable $J_o \subset J$. Consequently, $\lim_{n \rightarrow \infty} \mathbb{E}f(G_n) = \mathbb{E}f(G)$ for any bounded, Lipschitz continuous function $f : \mathbb{X} \rightarrow \mathbb{R}$. By the Portmanteau Theorem, this implies that $(G_n)_{n \geq 1}$ converges in distribution to G , and then the assertion of the theorem is a consequence of Lemma C.6, applied with $h(x) := 1 + \|x\|^2$. $\square$